Who Chooses Your Hospital?

Inherited Demand and the Measurement of Hospital Competition*

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August 30, 2026

Abstract

Hospital merger review relies on demand models in which patients choose hospitals. Using claims covering 8.5 million commercially insured Californians, I show that elective surgical demand is routed by referral menus that belong to primary-care organizations: menus barely move when an organization's physicians are replaced, transfer to physicians who change employer, and reallocate when organizations die. Demand is inherited, not chosen. A consideration-set model places 2.4 of 48 available hospitals on the effective menu. Replaying observed organization closures inside the models, the standard model recovers 41% of measured within-system diversion and the menu model 88%—a bias that clears mergers that should be flagged.

JEL: I11, L14, L22, L41, D43. **Keywords:** hospital competition, referrals, consideration sets, organization capital, merger review.

*Preliminary working paper; comments welcome.

1 Introduction

When a patient's primary-care practice closes, her next operation is likely to happen at a different hospital. If the practice she moves to already sends patients to her old hospital, she returns there 38.4% of the time. If the new practice has never sent anyone there, she returns 15.9% of the time. An otherwise similar patient whose practice does not change, measured on the same clock, returns 43.1% of the time.

She did not choose this. Neither did her physician: replace more than a quarter of a practice's doctors and its distribution of hospitals does not move at all. Nor, as far as I can measure, did anyone at the practice. The hospital her surgery ends up in was determined by an organizational habit that no one currently alive appears to have selected.

This happens constantly. Following California's commercially insured population from 2018 to 2023, I observe 3.06 million breaks in patient–practice attachment (closures, retirements, departures, plan changes), and only 14.6% of patients who have repeat surgery keep both their primary-care and specialist relationships intact across four years. The machinery that routes patients to hospitals is coming apart and re-forming continuously underneath the market.

And yet nothing moves. Within-county hospital market shares in 2018–19 and 2022–23 correlate at 0.979 (rank correlation 0.869; the median hospital's share moves 0.63 percentage points), across a window containing a pandemic. Hospital demand churns violently at the level of the individual patient and is almost perfectly still at the level of the market.

The puzzle is how that much turnover produces that little movement.

This paper's answer is that hospital demand is *inherited* rather than chosen. It attaches to organizational lineages of referring practices; it passes to new patients and to new physicians by succession; and when a lineage dies, its patients are not returned to the market but bequeathed to whichever organization succeeds it. Shares are stable not because the underlying relationships are stable—they are not—but because each broken link is replaced by one pointing the same way.

The object I estimate is the **effective menu**: the number of hospitals a patient's referral chain actually places in front of her, out of those available. A patient in my sample has 48.1 hospitals within reach that perform her procedure. Her chain puts **2.4** of them in front of her. That parameter, and the question of who chooses it, organizes everything that follows.

I use the California Health Care Payments Database, which covers essentially all commercially insured Californians: 8.47 million members, 184 million claims, and 5.1 million elective facility operations over six years. Three features make it possible to ask who routes demand. First, the data identify the rendering physician and the billing organization on every claim, so the referral chain (patient, primary-care practice, specialist

practice, facility) can be reconstructed rather than assumed. Second, the panel is long enough to observe practices dying and physicians moving. Third, because patients are followed across insurers, a relationship can be watched surviving a change of payer.

The identification rests on shocks to the referring organization that have nothing to do with hospitals. Practices close; physicians retire or leave the state; patients are administratively reassigned. I verify these events against federal provider records: National Provider Identifier deactivations, Medicare enrollment files, and the Doctors and Clinicians registry, so that a "closure" is an event with an external witness rather than an inference from missing claims. Throughout, results are reported separately for these forced severances and for voluntary switches, and the two are statistically indistinguishable. Four findings organize what the shocks reveal.

Nobody living chooses the menu. Two practices on the same street, treating patients with the same condition in the same year, send them to different hospitals: the dissimilarity between their hospital distributions is 0.78 against a measured sampling floor of 0.36. This is not geography. A rival practice in the *same five-digit ZIP code*, treating the same procedure group in the same year, predicts a patient's hospital at 0.184, against 0.126 for guessing the nearest capable facility, while her own practice predicts at 0.458. It is not the insurance network: build her practice's habit from only the patients it treats under *other* insurers, so that nothing about her own coverage enters, and it still names her hospital 0.395 of the time, 86% of what the full habit achieves. It is not price or contract design, and it is barely the physicians: turning over a quarter of a practice's doctors moves its hospital distribution about as much as turning over a tenth, and neither moves it more than a tenth of what being a different practice does. And a hospital cannot buy its way onto a menu: when a facility opens a procedure it has never offered, organizations that already send it other work adopt at 38% and organizations that do not adopt at 5%. Relationships do not merely hold demand in place; they determine who is able to compete for it.

The organization reprograms the physician, not the reverse. I follow 787 primary-care physicians who change employer, dating each move to the day of her first billing under the new organization. Before the move, 9.1% of her patients' operations occur at her future employer's leading hospital; afterward, 23.3%—a jump of 0.142 (s.e. 0.026, clustered on the destination organization)—while the share at her former employer's leading hospital falls from 21.3% to 8.7%. The two series cross at the quarter she arrives. Within one to three years she has closed 64 to 68% of the distance to a physician who was always there. The crossover survives among the patients she *retained* through the move, and a placebo arm of 3,791 non-moving physicians assigned a fictitious move date and destination shows no crossover at all.

Demand moves at succession, and along inheritance lines. Holding the type of surgery fixed, patients whose chain stayed intact change hospitals 29% of the time; when

the facility itself dies, 92%. Aggregating, at least 36% of same-service hospital switching is attributable to succession events rather than to drift, and this is a lower bound, because even "drifting" patients land on their current organization's leading hospital far more often than proximity would predict.

Standard demand models understate substitution inside systems. The policy-relevant question is where demand goes when a hospital becomes unavailable. I answer it twice, without a demand model. When referring organizations dissolve, the share of displaced demand retained inside the hospital's own system runs 1.4 to 2.8 times what availability implies, with bootstrap confidence intervals attached; when a hospital service line dies, 2.5 times. Two unrelated shocks, on opposite sides of the market, agree. The decisive comparison then replays the first shock inside the demand models—the same episodes, the same benchmark, the same ownership rule. Patients kept within the old facility's system 43.8% of the time; the patient-choice model predicts 18.1% and the menu model 38.8%. The standard model recovers 41% of measured within-system retention, below the measurement at every radius in every paired bootstrap draw; the menu model recovers 88%, with a residual shortfall that is itself statistically sharp, the pattern measurement error in the menus would produce. The correction is disciplined by measurement rather than by fit. In the same facility-death events, the referring organization's existing habit predicts where displaced demand lands far better than proximity does: mean absolute error 0.039 against 0.047, and it identifies the single largest destination 51.6% of the time against 23.4%.

Two consequences follow, and they are the reason to care about a parameter as small as 2.4.

Consumer-directed policy has nothing to grip. Two decades of cost-control policy assume a patient who shops: price transparency, high-deductible plans, shopping tools, reference pricing. The evidence that she does not is extensive; the usual explanations are behavioral. This paper offers a structural one: she is not the decision-maker, and the party that is inherited its habits and responds to no price signal I can find. Consistent with this, the patients who bear the most price risk exercise the least influence over price. Comparing the same person operating under a gatekept and an open insurance product, on 34,098 members observed under both, losing the chain's authority moves her to facilities 11.6 log points more expensive relative to her own alternatives (s.e. 0.004), while her own share of the bill rises roughly fivefold, from 3.2% to 16.2%.

Antitrust is watching the wrong layer. Concentration among referring organizations rose 25.9% between 2018 and 2023 in the same markets where hospital-system concentration rose 4.0%: six and a half times the rate, in the layer that actually allocates hospital demand. Because it happens through succession rather than transactions, no merger is filed and no screen sees it. Meanwhile the screens that do run inherit the understatement above: a demand model that misses the menu assigns displaced demand to the wrong

destinations, and does so in a direction that systematically understates how much of it a system keeps.

What the inheritance does *not* appear to do is protect patients. Using the exogenous arm, patients severed from their referral chain are no more likely to be readmitted within thirty days than matched patients whose chain was intact (-2.13 percentage points, s.e. 1.16; the ninety-five percent upper bound on harm is 0.15 points on a base of 5.4%). Whatever the menu encodes, it does not appear to be information about which hospital is right for this patient. The inheritance reallocates demand without reallocating health.

This paper connects three literatures.

To the **consideration-set** literature (Goeree, 2008; Gaynor, Propper and Seiler, 2016; Abaluck and Adams-Prassl, 2021), this paper contributes a new source of identifying variation and an uncomfortable finding. Existing work identifies limited consideration from advertising, from a policy reform, or from asymmetric price responses, and in each case an agent *chooses* the choice set: firms advertise, the state liberates, policy re-defaults. Here the choice set is assembled by a third party whose habits no living agent selected, and its *content* is identified from turnover in that party—variation available in any referral market, with no reform or price experiment required. The uncomfortable finding is about placement: on every measured reallocation event in these data, a model that loads the same routing information into utility performs as well as the consideration structure, so the data discipline *what* the choice model must carry rather than *where*. What survives as the consideration structure's own territory is the part no utility shifter can express: a countable menu, and revealed exclusions with error rates.

To the literature on **persistence and organization capital**, where Bronnenberg, Dubé and Gentzkow (2012) study brand capital and Klepper (2002) inherited capabilities, it offers a decomposition those papers could not run. Brand capital persists inside consumers, and migrants carry it. Here roughly half of what looks like consumer loyalty is not in the consumer at all: it does not travel with her, and it dies with the chain. Firms in this market inherit customers, not only capabilities, and the bequest is observable at the level of the individual transfer.

To **health-care industrial organization** it supplies the demand-side object that merger analysis assumes and does not observe. The workhorse tradition estimates hospital demand from patient-level choice among all capable facilities, with distance doing the identifying work (Capps, Dranove and Satterthwaite, 2003; Gowrisankaran, Nevo and Town, 2015; Ho and Lee, 2017); Ho and Pakes (2014) add what insurers pay physicians, identified from cross-plan comparisons. This paper keeps that machinery and relocates the identifying variation to turnover in the referring layer, to who died and who dissolved and who changed employer, and supplies two externally measured diversion ratios against which any demand model of this market can be checked. A separate strand studies how integration and incentives move referrals (Cooper et al., 2025; Whaley and Zhao, 2024;

Baker, Bundorf and Kessler, 2016; Brot-Goldberg and de Vaan, 2018; McCarthy and Richards-Shubik, 2025), identified, where it is causal, from acquisition and integration events. I study the stock those margins operate on, and find it nearly immovable.

The paper is closest in method to Finkelstein, Gentzkow and Williams (2016). They use patient migration to separate what belongs to the patient from what belongs to the place; here the identifying event is the organization's death rather than the patient's move, the logic Fadlon and Van Parys (2020) apply to physician exits in Medicare, and it separates what belongs to the patient from what belongs to the *firm that refers her*. The relevant "place" in health care turns out to be not a region but a lineage of referring organizations.

Section 2 describes the setting and the data: how a referral is made in California managed care, why the billing organization rather than the physician or the building is the unit of routing, the claims and registry sources the paper joins, and how the estimation sample is drawn. Section 3 establishes the premise the rest of the paper builds on: the menu belongs to the referring organization, and neither the map, the insurer's network, nor the physicians inside it explain it. Section 4 measures how much of a menu a patient inherits when her organization changes—in destinations, in the adjacent link of the chain, and in prices. Section 5 isolates the reduced form the model must reproduce: what an organizational break moves, and what it leaves alone. Section 6 writes the model that formalizes the premise; Section 7 its identification, including the tests that could have overturned it. Section 8 estimates the paper's named parameter, the effective menu of 2.4 hospitals out of 48—an object that stays between roughly 2 and 3.2 under every respecification tried, including facility fixed effects, while the taste corrections built on it do not—and tests the fitted menu against a pattern it was not fitted to. Section 9 delivers the punchline: measured against two real shocks, the demand model merger review relies on understates within-system diversion at every radius, the menu model tracks the measurement, and the layer that allocates hospital demand is consolidating with no transaction to review. Section 10 steps back from measurement and asks what kind of arrangement inherited demand is: for the patient inside it, for the firms that hold it, and for the markets that share its structure. Section 11 concludes. An online appendix collects construction detail and the robustness the main text compresses.

2 Setting and Data

This paper treats the referring *organization*—the medical group or foundation that bills for a patient's primary care—as the unit that routes demand. Three features of California managed care make that the right unit, and each is measurable in the claims. The section closes with the data and the sample.

2.1 The path to an operation

An elective operation in this setting is the end of a chain, not a decision. A patient with a complaint sees a primary-care physician. That physician refers her to a specialist. The specialist evaluates her, decides an operation is warranted, and performs it—or refers onward—at a facility where she holds admitting privileges or an ownership stake. Four distinct parties appear on the claims: the patient, the primary-care practice, the specialist practice, and the facility.

The specialist who receives the referral is usually not the one who operates. The first specialist a patient sees in the nine months before her operation performs it only 29% of the time; even the *last* specialist she sees—the pre-operative visit, a median of three weeks out—is the operating surgeon less than half the time (48%). The chain has four nodes, not three. This is why the object of interest is a practice's distribution over *facilities* rather than over surgeons: a referring physician has a usual hospital in a way that she does not have a usual surgeon.

The referral itself is never observed. California's payments database, like every all-payer claims database, records the `referring_provider_id` field as a sentinel value on essentially all lines; the referral is a clinical communication, not a billable event. What is observed is the *sequence*: an office visit with a primary-care physician, then a visit with a specialist, then an operation at a facility. Every statement in this paper about "routing" is a statement about that observed sequence, and I use the language of routing rather than of referral throughout to keep the distinction visible.

A related caution applies to the popular distinction between "gatekept" and "direct-access" care. In principle, health-maintenance products require a primary-care referral before a specialist visit will be covered, while preferred-provider products do not. In these data the distinction is a gradient rather than a switch: conditional on seeing both a primary-care physician and a specialist in the nine months before an operation, the probability that the primary-care visit came first runs from 0.49 for joint replacement to 0.69 for hernia repair across the seven pre-registered service lines. Nothing bypasses the primary-care physician, and even the least gatekept families route through her nearly half the time. The product indicator is therefore a coarse proxy for the chain's authority rather than a clean binary, and the two results that lean on it, the price-authority comparison of Section 8 and the override gradient of Section 10.1, are read as within-member contrasts across that gradient rather than as an experiment.

2.2 Why the organization is the unit

The unit of analysis in this paper is the billing organization: the entity that submits the claim, typically a medical group or a foundation, not the individual physician and not a physical office.

Not the physician. A physician's routing is largely inherited from whoever employs her. Two physicians in the same billing organization route alike whether they share an office or work twenty miles apart, while two physicians in different organizations at the same address do not. Measured directly on switches that cross an organizational boundary and switches that do not, the transfer of routing is 0.122 across against 0.043 within, a difference of 0.079 (s.e. 0.010), while within an organization the difference between sharing a site and not sharing one is 0.013 (s.e. 0.013), indistinguishable from zero. Granularity is bracketed from both sides: a coarser unit loses the boundary that matters and a finer one adds nothing. Section 3 shows the same thing dynamically: a physician who changes employer is two-thirds re-routed toward her new employer's hospitals within three years.

Not the building. Most billing organizations are not one place, which is why I write "organization" rather than "practice" wherever precision matters: the unit is an owner, and owners commonly span several buildings. The contrast just given settles which boundary matters: routing transfers across the owner and not across the address. It also means the paper cannot speak to *which* campus an episode used, since six address sources were tested and none recovers it, so multi-campus structure enters as a robustness consideration rather than as a fix.

Why the owner is the right level: the corporate practice of medicine. California Business and Professions Code §2052 bars corporations, including hospitals, from employing physicians directly to practice medicine. Vertical integration therefore runs through medical foundations and management-services organizations, which bill under their own identifiers rather than the hospital's. Two consequences follow. First, a hospital that has effectively acquired a physician group will usually not appear as the biller on that group's claims, so integration measured by billing identity misses most of it; this paper resolves ownership through Medicare enrollment reassignment and the American Hospital Association's system registry instead. Second, and more importantly for the unit choice, the legal structure makes the billing organization the natural locus of the employment relationship, the entity a physician joins, leaves and is paid by, which is precisely the boundary across which routing is observed to transfer.

2.3 Prices and who pays them

Three facts about money bound what any price-based explanation can do here. **Capitation is common and is not visible in prices.** Twenty-eight percent of service lines in this population are paid under capitation, rising to 53% within health-maintenance products. Capitated care is priced in the data only through the submitted-allowed field; the header allowed amount is zero on 17% of claims, and that zero is a contractual form rather than a missing value. Any analysis that filters on a positive allowed amount silently deletes most capitated care, a trap this paper avoids by constructing price from the submitted amount

where the allowed amount is structurally zero, and by counting the resulting unpriced share rather than imputing it.

The patient's marginal price is small and, for surgery, usually irrelevant. Cost-sharing rates average 3.2% of the facility's allowed amount in health-maintenance products and 16.2% in preferred-provider products. But elective surgery is expensive enough that most episodes exhaust the deductible and the out-of-pocket maximum: at the largest claim sizes the patient's marginal share of an additional dollar of facility payment falls to approximately zero. The one price margin patients demonstrably respond to is *network status*, whether the facility is in or out of network, which is a discrete, salient, and often catastrophic exposure rather than a marginal one.

Prices differ enormously across hospitals for the same operation. Standardizing by diagnosis-related-group weight to remove case mix, the ninetieth-percentile hospital is roughly twelve times more expensive than the tenth for the same procedure. Standardization changes the ranking and not just the scale: it pushes tertiary centers down and community hospitals up. A merger screen run on raw prices would attribute expensiveness to the hospitals treating the sickest patients.

2.4 What the plan controls

Insurers assign a primary-care physician of record to most enrollees, and it is tempting to treat that assignment as the relationship. It is not. Among members who saw any primary-care physician, the share who saw the *assigned* one is 0.627 in health-maintenance products and 0.184 in preferred-provider products, even though members in both see a primary-care physician at nearly the same rate (0.78 and 0.74). In preferred-provider products the assignment is paperwork. Throughout this paper the attachment is therefore *revealed*, constructed from the sequence of office visits, and the plan's record is used only as an independent witness against which the revealed measure can be validated.

Networks, by contrast, are real but stable. Over the six years studied I observe 39 transitions of a hospital from in-network to out-of-network, involving 36 hospitals and 12 payers and putting 3,447 episodes at risk. California commercial networks in this period did not churn. This matters twice: it rules out network rupture as a source of identifying variation, and it means the persistence documented in this paper is not an artifact of stable contracts being renegotiated into place.

2.5 Data sources and variables

The primary source is the California Health Care Payments Database, the state's all-payer claims database, maintained by the Department of Health Care Access and Information: every claim submitted for the state's commercially insured population, 2018–2023, with members followed under a single de-identified identifier across employers and insurers.

The paper uses four of its files: claim headers and service lines, which carry the rendering physician, the billing organization, the procedure, the site of care, and the payment fields; eligibility spells, which carry product type and the plan's assigned physician of record; and inpatient discharges, which carry the diagnosis-related group. The universe is members aged 18 to 64 with at least two ambulatory visits: 8.47 million people.

Five external sources are joined by provider identifier. The National Provider Identifier registry supplies each physician's identity and deactivation date; Medicare enrollment records supply the organization each physician reassigns billing to, which is how billing entities are resolved to owners; the federal Doctors and Clinicians registry supplies quarterly employment snapshots; the American Hospital Association's system compendium supplies hospital-system membership; and CMS hospital ratings supply the quality measures used in Section 10. Distances are computed between residential and facility ZIP centroids.

The constructed objects, each detailed in Appendix A, are: the *episode*, an elective facility operation with its professional and facility claims linked on member and date; the *referring organization*, the member's calendar-year modal biller among primary-care office visits; the *break*, a change in that attachment on a smoothed visit sequence, dated to the day and classified by verified cause; the *menu*, an organization's leave-one-out distribution over facilities; the *choice set*, the fifty nearest facilities within one hundred miles offering the episode's procedure group that year, with the chosen facility always included; facility *system ownership*, the modal system over the facility's panel years; and the *price index*, the facility's median payment for a procedure group relative to the statewide median for the same group, site and year, standardized by diagnosis-related-group weight for inpatient care.

2.6 The estimation sample

The estimation sample is commercially insured Californians aged 18 to 64, observed 2018–2023, undergoing elective facility-based operations. Across the full episode file, roughly half of operations occur in hospital outpatient departments, forty percent in freestanding ambulatory surgery centers, and ten percent as inpatient admissions, with median facility payments of about $3,400, $1,300, and $19,100 respectively; within the analysis frame, which additionally requires an elective admission and excludes obstetric and behavioral care, the inpatient share falls to three percent. Table 2 reports summary statistics for the frame and for the break spine.

Table 1 reports the construction, and Appendix A the episode and choice-set detail behind it. It separates two kinds of row, because they mean different things and reporting them together is how a funnel becomes a rhetorical device. Most rows are *population definitions*: an operation is excluded because it is outside the population the paper is about. Two rows are *resolution failures*: the operation belongs in the sample and the

pipeline could not attach what it needs. The larger of the two is the single biggest loss in the paper, at 18.9% of operations have no referring organization that can be identified, because the patient has no primary-care visits the claims can attach to her, or attends several organizations without a dominant one. That is a limit of the design, not a scope choice, and it is marked as such. It is also, on everything both groups can show, not a selection on routing: episodes with no identifiable referrer travel the same distances (median 7.8 miles against 8.0), sit inside the fifty-nearest choice set at the same rate (84.3% against 84.7%), land at facilities in the same price position, and use the nearest capable facility slightly *more* often (17.3% against 15.4%), so the routing facts below are not an artifact of conditioning on resolution. Where the two groups differ is composition, not conduct: the unresolved are more often inpatient (15.8% against 9.7%), slightly more often PPO, and resolution is within three points across products (82.9% HMO, 80.4% PPO) and essentially flat across years. Appendix B reports the full comparison.

One exclusion is applied two ways. Kaiser Permanente, which insures a large share of Californians, is a fully integrated system in which the insurer, the medical group and the hospitals are one organization. There is no arm's-length hospital choice to study inside it and no negotiated price to observe. Kaiser members appear in the database and are retained where they receive care outside the system.

The two ways of removing Kaiser are not equivalent, because a member may be insured elsewhere and treated at a Kaiser hospital, or the reverse. Excluding on the *payer* drops arm's-length operations that Kaiser members had at community hospitals; excluding on the *facility* keeps those and drops care delivered at Kaiser's own hospitals instead. They differ by 61,576 operations, about 1.2%. The reduced-form evidence uses the facility rule, which is the right one for a question about where patients are sent; the demand estimation sample uses the payer flag. I report both counts in Table 1 rather than a single "Kaiser excluded" line, and the difference is small enough that no result in the paper turns on which rule is applied.

Taken together, these features describe a market in which the patient rarely selects the hospital, faces almost no marginal price when she does, and is attached to a primary-care organization whose identity the insurer records imperfectly, while the organizations themselves are legally constituted as employers of physicians, span multiple buildings, and are paid in ways that make their own hospital choices nearly invisible in prices. It is a market built to make the question of this paper hard to see, and the reason it can be answered here is that all four parties appear on the claim.

3 Who Controls the Referral Menu

A patient who needs an elective operation has, on average, forty-eight facilities within reach that perform it. She is treated at one of them. Which one is set by a menu that

Table 1: Construction of the estimation sample

	Operations	Members
Panel A. The demand estimation sample		
Members in the HPD commercial universe (ages 18–64, ≥ 2 ambulatory visits)	—	8,471,732
Elective facility operations, 2018–2023	5,068,724	2,712,742
excluding Kaiser-paid episodes	4,804,081	2,546,770
facility resolved and in California	4,730,997	2,513,091
referring organization identified†	3,838,612	2,062,153
patient locatable and choice set built†	3,805,151	2,048,573
Demand estimation sample	**3,805,151**	**2,048,573**
episode × candidate rows	183,179,267	—
of which the seven pre-registered service lines	232,363	211,228
Panel B. The identifying variation		
	Breaks	Members
Primary-care breaks on the smoothed sequence	3,057,664	2,049,577
both organizations above the volume gate	2,796,850	1,909,775
exact residential ZIP held across the break	2,381,034	1,685,883
Mover panel (an episode on both sides)	**84,569**	**84,375**
episodes on those breaks	308,706	—

Notes: † Rows marked with a dagger are *resolution failures*, not scope decisions: the operation belongs in the sample and the pipeline could not attach a referring organization to it, or could not place the patient. Every other row is a population definition. The single largest loss in the paper is the first dagger, at 18.9% of operations.

Panel A excludes Kaiser by the *payer* flag, which is what the demand sample uses; the reduced-form evidence elsewhere in the paper excludes Kaiser *delivery facilities* instead. The two rules differ by 61,576 operations — the facility rule keeps arm's-length care at community hospitals that the payer flag drops, and drops care at Kaiser's own hospitals that the payer flag keeps.

Panel B is drawn entirely from the mover-panel cascade so that its steps are mutually consistent; it is not the same filter sequence as Panel A.

belongs to the organization that refers her, rather than to her, to her insurer, or to the physicians inside it. That menu is narrow to begin with: the median referring organization sends 41.9% of its patients' operations to a single hospital. The evidence in this section uses no demand model.

3.1 Not the map, not the network

Begin with the simplest possible question. Given a patient's episode, how often does a rule correctly name the facility she actually used?

Table 3 scores four such rules on the same 3.73 million episodes. The own-organization rules are computed leaving her own episodes out of the history they are built from, so

Table 2: Summary statistics

Panel A. In-frame elective operations	
Operations	4,418,007
Members	2,367,753
hospital outpatient department	51.4%
freestanding ambulatory surgery center	45.2%
inpatient admission	3.4%
Seven pre-registered service lines	5.9%
HMO at the episode	36.7%
PPO at the episode	52.9%
Referring organization resolved	82.4%
Specialist organization resolved	85.4%
Median miles to the chosen facility	8.4
Treated at the nearest facility in the set	15.7%
Mean capable facilities in the choice set	48.2
Panel B. Primary-care breaks	
Breaks	3,057,664
Members	2,049,577
practice closed	1.6%
physician left the market	1.2%
destination newly acquired	1.1%
physician changed employer	12.6%
Return to a previous organization	12.5%
Median days between organizations	210

Notes: Panel A describes the in-frame elective operations of Table 1; specialist resolution is computed outside the 2018 lookback-censored window. Distance and choice-set size come from the chosen row of the episode's candidate set (the fifty nearest facilities within one hundred miles offering the procedure group that year, with the chosen facility always included). Panel B describes every break on the smoothed primary-care attachment sequence; cause shares are not exclusive and the remainder is unexplained. The nearest-facility share is computed on the demand sample; the reduced-form frame's share, quoted in Section 3, is 14.9%.

she cannot predict herself. The twin is a different organization that also treats patients living in the same five-digit ZIP code, for the same procedure group in the same year—the tightest same-neighborhood comparison the data support, and one that retains 99.7% of episodes.

It is not geography. A rival organization on the same streets, seeing the same kinds of patients for the same kinds of operations, predicts at 18.4% against a map's 12.6%. Her own organization predicts at 45.8%. Whatever the referring organization contributes, two and a half times what a neighbor contributes, it is not a repackaging of distance.

One version of the comparison needs no geographic matching at all, because it is made *within* a single episode. Consider the facility a patient actually used. Her own organization sends 32.4% of its volume there; the twin sends 12.5%. Now consider the facilities in the

Table 3: Predicting the facility a patient actually used

Rule	Names it exactly	In its top three
Go to the nearest capable facility	12.6%	30.4%
Go to a twin organization's leading hospital	18.4%	35.2%
Go to **her own** organization's leading hospital	**45.8**%	**68.8**%
Go to **her own** organization's leading hospital, built only from its patients on other insurers	39.5%	60.0%

Notes: All four rules are scored on the same 3.73 million elective operations, 2018–2023. "Names it exactly" is the share of episodes where the rule's top-ranked facility is the one used; "in its top three" allows the used facility to be any of the rule's three highest-ranked. Own-organization rules leave the focal episode out of the history they are built from. A twin is a different organization that also treats patients *living in* the same five-digit ZIP code, for the same procedure group in the same year. The final row is undefined for the 7.9% of episodes at organizations with no other-payer patients, which are excluded from its score rather than counted as failures.

same choice set that she did *not* use. Her own organization sends 1.0% of its volume to them; the twin sends 1.3%—if anything, slightly more. The two organizations are scored on the same facilities, for the same patient, in the same year. The own-organization advantage exists at the hospital she was sent to and nowhere else: shared geography would raise her organization's share at every nearby facility, and it does not.

A coarser fact points the same way. Only 14.9% of patients are operated on at the nearest facility capable of performing their procedure (12.6% under Table 3's stricter exact-name scoring); the median patient passes several closer hospitals that could have done her operation.

It is not the insurance network. The obvious alternative is that the menu is the insurer's list of covered hospitals under another name. The third row settles it. Build her organization's habit from *only* the patients it treats under *other* insurers—so that nothing about her own coverage enters the prediction—and it still names her facility 39.5% of the time, 86% of what the full habit achieves. An organization's routing survives dropping the insurer almost intact.

Comparing the hospital distributions themselves, rather than scoring predictions from them, gives the same answer: the insurer accounts for between a quarter and a third of the difference between two organizations. Networks matter. They are not the phenomenon.

3.2 The organization reprograms the physician

If the menu were the aggregation of its physicians' preferences, it would belong to them and travel with them. Two designs say it does not: one follows physicians who change employer, the other watches organizations that replace their staff. They answer the same question from opposite sides, and the first is the sharper of the two because it can say which way transmission runs.

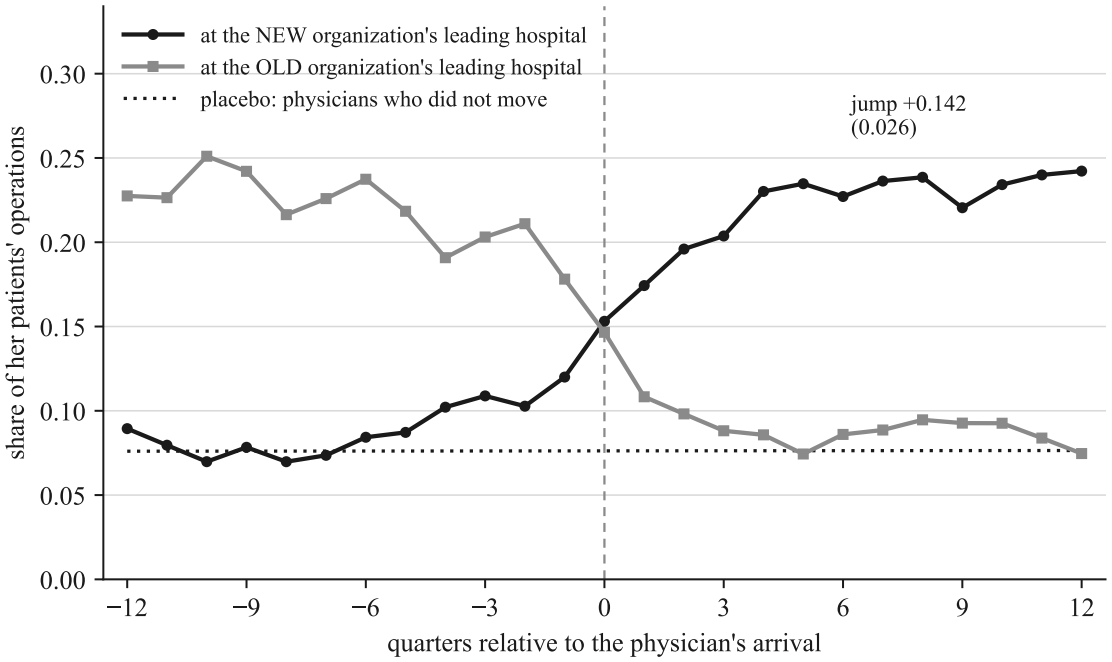


Figure 1: The organization reprograms the physician

Notes: Share of a moving physician's patients operated at her new organization's leading hospital and at her old one's, by quarter relative to the day she first bills under the new organization. Menus are computed leaving her own patients out, so a rise is not mechanical. The placebo line is the same quantity for physicians who did not move, assigned a fabricated move date and a fabricated destination drawn from same-market organizations. Sample: 787 arm's-length movers; the jump is clustered on the destination organization. The 787 are drawn from 2,455 raw changes of billing organization, most of which are not moves—one firm relabeling itself, or a shift between arms of the integrated system whose facilities are outside this sample—so an arm's-length change of owner is required.

I follow 787 primary-care physicians who move between arms-length organizations, dating each move to the day she first bills under the new one. Menus are computed leave-this-physician-out, so her own patients cannot define the benchmark she is scored against.

Figure 1 shows what happens. Before the move, 9.1% of her patients' operations occur at her future employer's leading hospital; afterwards, 23.3%. Over the same window the share at her former employer's leading hospital falls from 21.3% to 8.7%. The two series cross in the quarter she arrives. The estimate behind that sentence is the arrival jump γ in

$$y_{p\tau} \;=\; \alpha_p \;+\; \gamma\,\mathbf{1}\{\tau \geq 0\} \;+\; \varepsilon_{p\tau}, \tag{1}$$

where $y_{p\tau}$ is the share of physician p's retained patients' operations at the destination organization's leading hospital in event quarter τ, estimated within physician on balanced windows. The jump is $\hat{\gamma} = 0.142$ with a standard error of 0.026, clustered on the destination

organization, which is the level the design requires: the physicians in this arm share only 211 destination organizations between them, and standard errors clustered on the physician rather than on the destination would be roughly forty percent smaller.

Within one to three years she has closed 64 to 68% of the distance between her old routing and that of a physician who was always at the new organization, with the benchmark computed separately for each cell rather than borrowed across them. The residue—roughly a third—is hers. Routing follows the employer, not the physician.

The crossover is not her patient list turning over. A physician who moves acquires a new book of patients, and if those patients simply live nearer the new employer's hospital the crossover would be mechanical rather than behavioral. Splitting her post-move episodes settles it. Among the patients she carried with her, 22.7% of operations occur at the new employer's leading hospital; among patients new to her, 25.6%. She routes to the new employer at nearly the same rate for both groups, so the rise is not her patient list turning over.

The two groups differ on the other side. Her carried patients still use the *old* employer's leading hospital 20.1% of the time; her new patients only 10.3%. She adopts the new employer's hospitals for everyone, but the patients who came with her hold on to the old one. Both a patient's own history and her organization's are at work, which is why the model of Section 6 carries two of them.

The move need not be exogenous. Physicians who change employer may differ from those who do not, and may change for reasons connected to where they want to send patients. What the design uses is the *timing*: the pre-period is flat for three years, the break is sharp at the quarter of arrival, and a placebo arm of 3,791 physicians who never moved, assigned fictitious move dates and destinations, shows no crossover at all. A selection story would have to explain a discontinuity at a date the physician's employer, not the physician, determines.

The same question, asked of organizations rather than people. The mover design uses only physicians who change employer, and they may be unusual. A second design uses every organization instead. If physicians carried the menu with them, an organization that replaced many of its physicians should route differently afterwards.

I measure an organization's hospital distribution in consecutive years and compare the movement against a size-matched split-half floor computed on the same organization-years, so that sampling noise is differenced out rather than assumed away. Year-to-year drift above that floor is 0.036. For scale, the excess dissimilarity between two *different* organizations in the same market is 0.42—twelve times larger. Menus are stocks, not flows.

Figure 2 decomposes that drift by how much of an organization's staff turned over, where turnover is the share of evaluation-and-management volume billed by physicians new to the organization.

Two things are true here. Hiring at all does move the menu: drift rises from 0.029

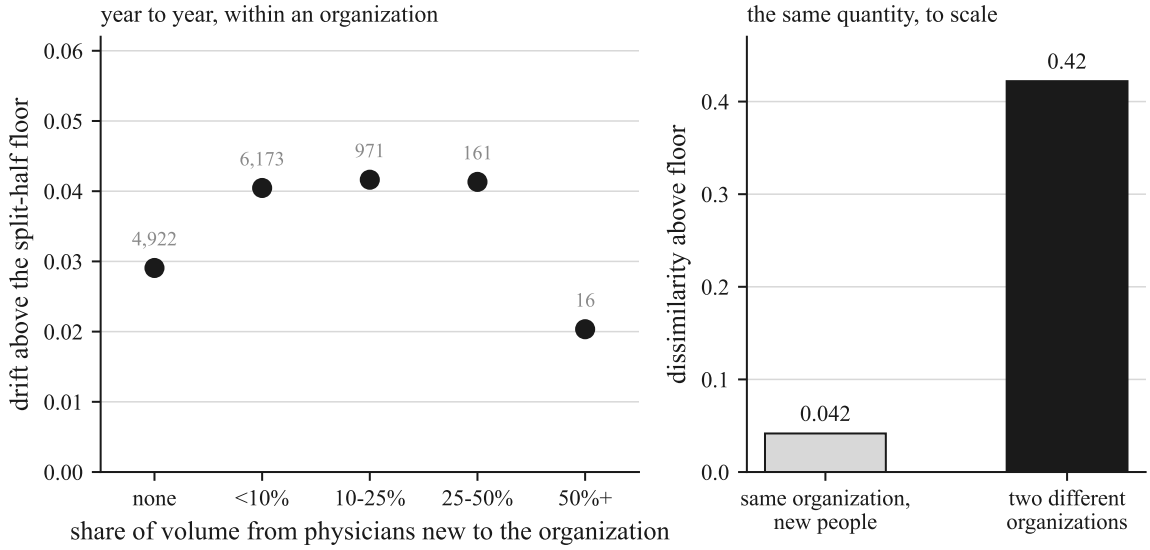


Figure 2: Replacing the people does not move the menu

Notes: Left panel: an organization's year-to-year drift in its hospital distribution, above a size-matched split-half floor computed on the same organization-years, by the share of its evaluation-and-management volume billed by physicians new to it. Organization-year counts appear above each point. Right panel: the largest of those cells against the dissimilarity between two *different* organizations in the same market, both measured as excess over their own floors. Note the difference in vertical scale between the panels.

where no volume comes from new physicians to about 0.041 where some does. But the *intensity* of hiring does not matter—0.040, 0.042 and 0.041 across the three cells that span a tenth to a half of an organization's volume. And the whole range, from 0.029 to 0.042, is at most a tenth of the 0.42 that separates two *different* organizations in the same market.

Two limits. The cell above half of volume contains sixteen organization-years, so wholesale replacement is not tested here—the flat-intensity claim rests on the three middle cells. And without the size-matched floor the relationship inverts, because small organizations have both noisy menus and few entrants, which is why the floor is computed on the same cells rather than pooled.

Both designs are about the *distribution* of hospitals, not about clinical practice: physicians certainly bring their own treatment styles with them, as Fadlon and Van Parys (2020) show. What they do not bring is the hospital.

4 The Movers Design

Section 3 establishes that referral menus are organizational and that they survive the people inside the organization. What it does not measure is how much of a menu a patient inherits. This section does, using patients whose referring organization changes while they

themselves do not move, and asks a single quantity of them: how much of the difference between two organizations travels with the patient.

4.1 Design and identification

Let i index a break, a patient whose referring organization changes from $A(i)$ to $B(i)$ while she herself does not move. Let j index a destination, the hospital at which an operation happens, let $r \in \{A(i), B(i)\}$ index one of her two organizations, and let h denote the facility a given episode used. Write λ_{rj} for organization r's true propensity to route toward j, and let y_{ijt} indicate that patient i's episode in period t went to j. The two periods are the calendar year before the break and the year of it, so $r(i,t)$ is A before and B after, and

$$y_{ijt} \;=\; a_{ij} \;+\; a_t \;+\; \delta\, \lambda_{r(i,t)j} \;+\; \varepsilon_{ijt}, \tag{2}$$

in which δ is the share of the difference between two organizations that binds a continuing patient within the horizon studied.

The fixed effect is on the patient–destination pair, and that is the design rather than a detail. It absorbs everything permanent about this patient at this destination—how far away it is, whether she has been treated there before, whether her family uses it—so the only thing left to identify δ is what changed at the break. A patient-level effect would not do this: it removes her propensity to be operated on at all and leaves her propensity toward this destination, which is exactly what an organization's habit is correlated with.

Those propensities are not observed. What is observed is where each organization's other patients went. Write $s_r(j)$ for the share of organization r's patients' episodes that go to j, measured for both organizations over the same five calendar years centered on her break and built from members who never switch. That last choice decides what the equation measures: were s_r to count the focal patient's own episodes, her pre-break episode would enter $s_{A(i)}$ at exactly the destination she is about to leave, and she would predict herself. The common window is what makes the comparison clean. Write the propensity actually observed in year t as $\lambda^t_{rj} = \lambda_{rj} + m_{tj}$: the organizational part of (2) plus a market-wide part belonging to that year. Differencing two organizations observed over the same span removes m_{tj} exactly, so a market-wide shift in where patients are sent cannot enter the regressor; differencing across spans leaves it there. The width of the window trades precision against recency and is not an identifying choice; Appendix D reports the alternatives. Substituting $s_{r(i,t)}(j)$ for $\lambda_{r(i,t)j}$ in (2) gives the equation taken to the data. With two periods its within estimator is algebraically a first difference of the patient's destination on the gap

$$G_{ij} \;=\; s_{B(i)}(j) \;-\; s_{A(i)}(j), \tag{3}$$

and returns the identical coefficient; the panel is written out because it is the form the fixed effects and the event study extend.

Equation (2) has an event-study representation, and that is the form the section reports. When origins and destinations both vary, plotting the raw outcome around a move shows nothing, since moves in opposite directions cancel; Finkelstein, Gentzkow and Williams (2016), following Bronnenberg, Dubé and Gentzkow (2012), rescale each mover's outcome so that 0 is her origin's average and 1 is her destination's, at which point the jump is the parameter. Their outcome is a scalar, log utilization, so the gap between origin and destination is a number to divide by; ours is a choice among hospitals and has to be made scalar first. By (3), G_{ij} is positive where the new organization sends more of its patients than the old one, so evaluating it at the destination h an episode actually used scores that episode high when it looks like the new organization's routing and low when it looks like the old one's. What that score averages under each organization's own distribution,

$$E_i^A \;=\; \sum_j s_{A(i)}(j)\, G_{ij}, \qquad E_i^B \;=\; \sum_j s_{B(i)}(j)\, G_{ij},$$

plays the part of the origin and destination means, and the difference between them is $\sum_j G_{ij}^2$, positive whenever the two organizations route differently at all. Rescaling gives

$$\tilde{\delta}_{ih} \;=\; \frac{G_{ih} - E_i^A}{E_i^B - E_i^A}, \tag{4}$$

which averages 0 over episodes routed like the old organization and 1 over episodes routed like the new one, and whose jump at the break is δ. A single operation is a coarse draw, so individual values fall outside that interval; the mean is the object of interest.

The two forms need different candidate sets. As a per-episode *level*, (4) needs no truncation, since each episode contributes one number however long the two organizations' destination lists are, and their union runs to a median of 294 facilities. As a regression *slope*, (2) puts one observation on every candidate, so the set has to be finite, and it is the ten facilities each organization sends most of its patients to. Which one is used turns out not to matter, and Appendix D sweeps it. The overlap between the two organizations does matter: both use only about a fifth of the union, on the ten-facility sets and on the full lists alike, so most of what the design reads is places one organization sends patients to and the other does not.

Least squares recovers δ only if whatever else changed at the break is unrelated to this particular pair of organizations' gap, and two things can break that. The first is sorting: a patient who chooses a new organization may choose it partly for where it sends people, or a health shock may prompt both the switch and a new destination. The design answers this in two places. Inside the estimator, the patient–destination fixed effect removes the

standing reason a patient and her organization agree about a destination, which is that both are near it. Across samples, it compares her against patients whose organization dissolved or whose physician left the market—events verified against federal provider records, and not plausibly chosen. If the estimate rested on patients selecting organizations for their routing, the forced arm would be smaller than the voluntary one at a horizon long enough for the two to be comparable—a qualification that matters, because a dissolved organization has no referral pipeline and its patients' operations therefore arrive on a different schedule.

The second is measurement, and it bounds how the estimates may be read rather than threatening the first condition. The gap is estimated from finite habit pools, and classical measurement error in a regressor attenuates the slope toward zero whether or not anything is sorting. Every estimate below is therefore a floor until Section 4.2 measures the error and lifts one of them.

Three placebo designs ask whether the coefficient reads the gap or some property of switchers, each run with the identical estimator. The first assigns a non-switching patient a fabricated break date and a destination organization drawn from the empirical distribution: if the estimator responds to the mere passage of time in a panel, it fires here. The second pairs each patient's movement with a *different* patient's organizational gap, which must return zero if δ measures the gap rather than the switcher. The third estimates the same equation between the two years *before* the break, where nothing has yet happened.

The design has a name in another literature. The organizational share $s_r(j)$ is a leave-one-out estimate of a unit's latent effect used as a regressor, which is how teacher value-added is constructed (Kane and Staiger, 2008; Chetty, Friedman and Rockoff, 2014), and the equation asks of it what that literature asks: not whether the share is exogenous—no average of realized choices is—but whether it *forecasts*. Chetty, Friedman and Rockoff (2014) answer with teachers who switch schools; the switching patients here are the same test, with δ in the role of the forecast coefficient, and the placebos above in the role of their quasi-experimental controls. Two differences keep the analogy honest. Their coefficient of one is the benchmark of full transfer, while δ is a share transferred *within a horizon*, so a value below one mixes forecast quality with the speed and completeness of inheritance and is not a bias estimate. And their remaining check—that variables excluded from the value-added model do not predict its target—has an analog here with one twist. In their setting the excluded variable (parent income) is generated outside the assignment being tested; a patient's own pre-break history is generated by her *previous* organization's menu, and neighboring menus overlap, so its correlation with the new menu cannot be zero even with no sorting at all. The test is therefore a difference rather than a level: the correlation where she chose the destination organization, against the correlation where a closing practice's roster settled it for her. Sorting lives only in the gap between the two, and it is reported with the results. Finkelstein, Gentzkow and Williams (2016) is the same

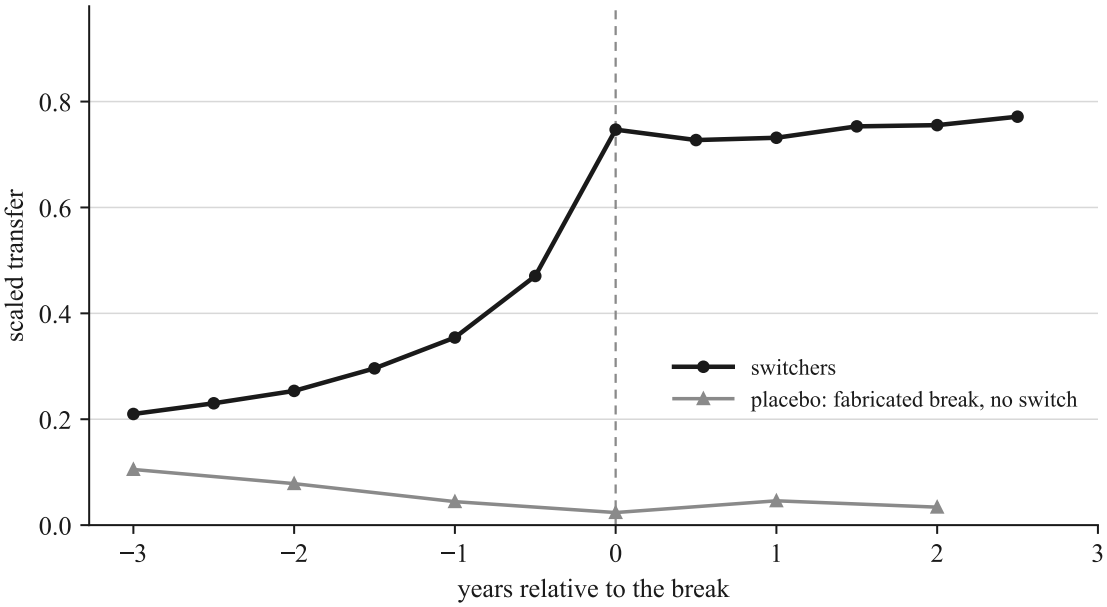


Figure 3: Routing transfers at the break

Notes: The outcome is scaled so that 0 is the old organization's hospital distribution and 1 is the new one's; the vertical line is the break. The placebo assigns a fabricated break date to patients who did not switch and runs the identical estimator. The two series are binned differently—half-years for switchers, whole years for the placebo, as each was estimated—so they are plotted on a common axis in years at their bins' left edges, and the placebo's markers are twice as far apart in underlying time. Bins hold between 12,185 and 108,972 episodes.

logic applied to places rather than organizations.

4.2 Results

Figure 3 plots (4) by time relative to the break, with hospitals as destinations. Three years out the patient's episodes sit near her old organization's hospital distribution; they drift upward as the break approaches, jump at it, and are flat within a half-year. The placebo series—patients who never switched, given a fabricated break date—is near zero throughout. The drift before the break is anticipation rather than a failure of the design: patients begin visiting the new organization before the old attachment formally ends, so episodes in the last months before the break are already partly referrals from the new one.

The level requires one decision, and it used to be a free parameter. Episodes close to the break are ambiguous—a referral made under the old organization can be operated under the new one—so a window around the break is conventionally excluded. Sweeping that window from zero to two years moves the estimate monotonically from 0.379 to 0.518 with no interior optimum, which means the analyst chooses the answer. I replace the clock with the event. The baseline period is defined as the episodes occurring before the patient's *first visit* to the new organization, which is when drift toward it can begin.

This requires no window to be chosen, and it gives $\hat{\delta} = 0.497$, against 0.480 under the conventional twelve-month exclusion. The convention was close to right; it is now justified rather than assumed.

About half of the difference between two organizations' hospital distributions travels with a patient who moves between them, holding her address fixed. Read as a decomposition, half of what looks like a difference between two organizations is a difference in where they route people and the rest stays with the patient—and since noise in the measured menus attenuates the estimate, half is a lower bound on the organization's share. Organizations at the tenth and ninetieth percentile of the share of patients they send to a given hospital differ by 6.9 points in that share, and a patient who moves between two such organizations takes 3.4 points of that difference with her. For scale, the median organization that uses a hospital at all sends 0.7% of its patients there.

The level cannot be corrected for measurement error, because a reliability corrects a slope and a level is not one. The same design estimated as a slope can be, and it is worth doing: it says how far below the truth the level sits. On the 69,252 switchers the slope form gives 0.260 (0.009), clustered on the destination organization. The two are one design computed two ways, and almost all of the distance between 0.497 and 0.260 is the baseline period rather than the estimator. Recomputing the level on the slope's window—the calendar year before the break, in place of a baseline dated from the patient's first visit to the new organization—gives 0.314, while restricting it to the slope's candidate set moves it by 0.004 at any width (Appendix D). The year before the break is where the anticipation of Figure 3 sits, so a baseline drawn there is measured partly after the drift toward the new organization has begun. What is left between 0.314 and 0.260 is the attenuation this section measures, and it is the right size: correcting the slope for noise in the menus returns 0.300.

Every refinement of the sample moves the slope up, which is what attenuation would produce—though not only attenuation, as the measured correction below shows. Restricting to decisive switches rather than single-visit ones gives 0.371; to breaks where two independent definitions of the switch agree on both organizations, 0.302; and to switches accompanied by a plan reassignment, 0.417. The converse cells behave as they should: where the plan contradicts the switch the transfer is 0.227, and where the switch rests on a single visit *and* the plan contradicts it, 0.211. The two margins interact as the institutions predict, and the interaction is sharper where the organization holds gatekeeping authority (Appendix D).

The size of the attenuation can be measured rather than inferred from that ladder. Split the pool of never-switching patients in two and build each organization's distribution twice on disjoint halves: the two measures carry the same signal with independent errors, they correlate at 0.785, and by the Spearman–Brown relation the menus used above are about 88% signal. Correcting for it gives 0.300 (0.009). Three further routes agree without being

made to, and they are algebraically different estimators, so the agreement is a test rather than a restatement: the most conservative of them is an errors-in-variables correction using the menus' exact multinomial sampling variance, which gives 0.285 because it credits them with only the noise independent sampling would produce, and the noise actually present is two thirds larger again. Appendix D gives all four, their over-identifying checks, and the estimator's two free constants, across which the corrected estimate moves by 0.002 and the uncorrected one by six times that.

So the slope is understated by about 16%. Every cell of the lattice sits at or above the corrected figure rather than scattered around it, which says the refinements do more than remove noise: a decisive switch, or one an insurer independently confirms, is not merely better measured but a larger treatment. That is why the correction is measured rather than read off the ladder. The level is built on the same menus and cannot be exempt from the same attenuation, so 0.497 is a floor, and one of roughly known depth.

None of the three designs of Section 4.1 returns the effect. The fabricated break gives -0.091, opposite in sign to the real effect and about a fifth of it in magnitude. The permuted pair gives 0.0004 (s.e. 0.0006), indistinguishable from zero. The pre-trend gives 0.003 (s.e. 0.010). Table 4 gives their intervals and sample sizes.

The forced arm behaves as the design requires. At a three-year horizon the two arms are the same, and the difference contains zero at both the member and the destination-organization clustering levels—separately for physician exits and practice closures, not only when they are pooled. On the sharpest cell available, breaks where an independent source confirms the switch was unchosen ($n = 881$), the transfer is 0.448 and the difference from voluntary again contains zero.

The horizon explains an apparent difference between the two arms. At one year forced movers transfer *less* than voluntary ones (0.391 against 0.462, a gap of 0.071); by two years the gap is 0.026 and by three it is gone. Read as selection this is backwards—selection does not expire. Read as take-up it is mechanical: a dissolved organization has no referral pipeline, so its patients' operations cannot be its in-flight referrals, and the two arms converge as the backlog clears. Selection is nonetheless visible, in the pre-period rather than the level: voluntary switchers drift toward their future organization before the break by $+0.130$, while closures drift by -0.084. That is the anticipation a chosen switch should produce and an unchosen one should not.

One forced cell does drift—patients whose physician left the market anticipate by $+0.231$—and the explanation is dating rather than selection: in 62.8% of those breaks the physician had already stopped billing before the recorded break date, so the pre-period contains post-treatment time, and re-dating the event to his last active day collapses the anticipation to 0.072. Appendix D gives the re-dating and the ordering across exit types that confirms it.

Which fixed effect does the work can be shown rather than argued, and Table 5 reports

Table 4: The movers design: main estimates and placebos

	Estimate	95% CI	Movers
Pass-through at the break, equation (4)			
Event-based baseline	0.497	—	
Conventional twelve-month exclusion	0.480	—	
Window sweep, 0 to 730 days	0.379–0.518	—	
Chosen and unchosen breaks			
Voluntary	0.430	$[0.398, 0.466]$	67,613
Forced: exit and closure	0.432	$[0.370, 0.488]$	5,585
difference, member-clustered	—	$[-0.066, +0.066]$	
difference, practice-clustered	—	$[-0.126, +0.116]$	
Placebo: fabricated break, non-switchers	-0.091	$[-0.101, -0.079]$	180,938
Placebos on the slope, equation (2)			
Permuted referring pair	0.0004	$[-0.0009, +0.0016]$	68,882
Pre-trend, $t-2$ to $t-1$	0.0026	$[-0.0161, +0.0213]$	17,218

Notes: Every destination is a hospital. The first two panels are the scaled transfer of equation (4), in which 0 is the old organization's hospital distribution and 1 is the new one's; the third is the slope of equation (2), a probability per unit of gap. The two scales are not comparable and the panels should not be read across. Intervals in the first two panels are member-block bootstraps; in the third they are ± 1.96 times a standard error clustered on the destination organization. The event-based baseline replaces the exclusion-window choice with the patient's first visit to the new organization, and the sweep row brackets the specification margin that choice removes; those three rows share one pooled sample of breaks and differ only in how the baseline period is drawn, so no mover count distinguishes them, and that baseline holds 192,357 episodes. The voluntary arm is the set of breaks at which no cause flag fired, so it is a residual and not a stated choice. Counts are movers for the transfer rows, fabricated movers for the non-switcher placebo, and switchers contributing to the regression for the slope placebos.

every available effect. Destination-by-period effects absorb a shock to a hospital in the year of the break—a service line opening, a unit closing. Patient-by-period effects absorb everything about the patient at that moment: how many operations she had that year, a change of plan, a health shock. Those are estimable only because she faces several candidate facilities in each period, so the regressor varies inside the cell rather than being absorbed by it. None of them moves the coefficient by as much as 0.02. Against that, replacing the patient–destination effect with a patient effect moves it by 0.537, more than twice the coefficient itself. That gap is the geography confound priced directly, and it is why the design controls not for the patient but for the patient at this destination.

The gap is fixed within a patient–destination cell, so (2) extends to an event study by interacting it with time since the break and omitting the year before it. Figure 4 plots the path. Read across all cells the leads are not zero, which would be an anticipation problem if it were what it looks like. It is composition: the year on either side of the break carries over a million observations while the third year back carries a fraction of that, and the

Table 5: The transfer under every available fixed effect

	(1)	(2)	(3)	(4)	(5)
Organization's habit $s_{r(i,t)}(j)$	0.797	0.260	0.257	0.271	0.266
	(0.012)	(0.009)	(0.009)	(0.009)	(0.009)
Patient	Y				
Patient × destination		Y	Y	Y	Y
Destination × period			Y		
Patient × period				Y	
Origin organization × period					Y

Notes: Fixed effects included in (2) are marked below the coefficient; column (2) is the specification reported throughout. Standard errors, in parentheses, are clustered on the destination organization; all five round to 0.009 apart from column (1)'s 0.012. Column (1) is not a robustness check but a counter-example: a patient effect removes her propensity to be referred at all and leaves her propensity toward *this* destination, which is what the habit is correlated with. The origin-organization effects of column (5) are shown against the patient–destination effect alone, because each break has one origin organization and they are otherwise nested inside the patient-by-period effects of column (4).

cells surviving that far are patients with long panels and frequent operations—a different population, not an earlier one. Restricted to cells present at every horizon, the lead is 0.003 (s.e. 0.013). That is the pre-trend placebo of Table 4 drawn as a path, on a different sample and by different machinery, and it returns the same 0.003.

After the break the transfer drifts down, from 0.260 to 0.208 by the third year, and the decline is entirely the treatment ending rather than the relationship fading. Patients average 1.49 breaks, so a growing share have left the arriving organization for a third one by the second year. Restricting to breaks with no later switch—which conditions on the future and is therefore a diagnostic rather than an estimate—the path does not decline at all: it *rises*, 0.272, 0.286, 0.303, 0.312. What decays is exposure to the menu, not the hold the menu has.

The additive structure of (2) implies that transfer is a property of the *gap* between two organizations rather than of the organizations themselves: a patient moving from A to B and a patient moving from B to A should transfer the same share in opposite directions. This is one of the checks Finkelstein, Gentzkow and Williams (2016) apply to their own mover design, and it is testable only on pairs observed in both directions, of which there are 8,694. The transfer is 0.3588 in each direction, on samples that share no patients. The two agree more closely than 8,694 pairs could warrant, so the last digits are luck; what the test asks is whether the two directions differ materially, and they do not.

The panel form asks the same question a second way, and answers it less comfortably. The arriving organization's habit enters after the break and the departing one's before it, and the difference form makes them share a single coefficient, equal and opposite. Estimated separately they are 0.269 and 0.250, and a Wald test of equality returns

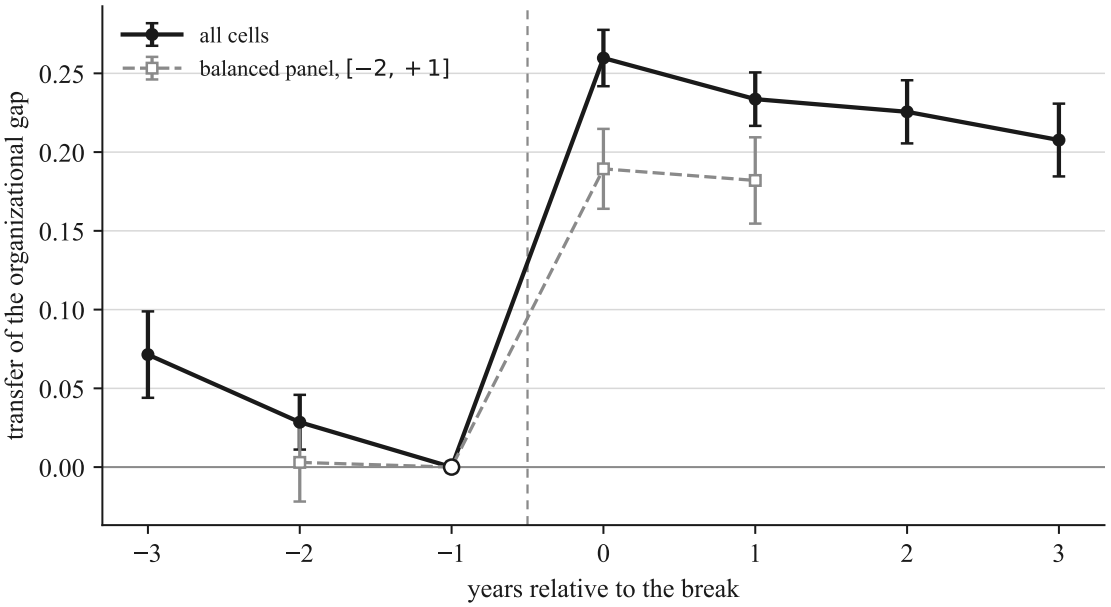


Figure 4: The transfer by year relative to the break

Notes: Coefficients on the organizational gap interacted with year relative to the break, from (2) with patient × destination effects; bars are 95% confidence intervals clustered on the destination organization. The year before the break is the omitted reference, drawn hollow at zero, and the dashed line marks the break itself. The balanced series holds the set of patient–destination cells fixed across $[-2, +1]$, so its levels are not comparable to the other's—patients operated on often enough to appear at every horizon transfer less—but its leads are the ones to read.

$\chi^2(1) = 5.20$, so on more than a million observations the restriction is rejected—barely, at $p = 0.02$, and by about seven per cent of the coefficient. The obvious mechanical explanation does not account for it. Organizations patients leave are dying by construction and are measured from thinner pools—a median of 235 patient-years against 317 for the organizations they join—which would attenuate the departing side more. But restricting to breaks whose two pools are within a quarter of each other, where that asymmetry is absent, leaves the gap if anything larger (+0.025 against +0.018). What survives is small and points the way the menu does: the arriving organization's habit binds slightly harder than the departing one's releases, which is what a menu operating by inclusion should produce.

Organizational identity here is a billing entity, and one firm can hold several. Sorting breaks by how far the patient actually moved organizationally gives the ladder of Table 6, whose bottom rung is an internal placebo the design did not have to pass: a second billing identity at the same address, with the same physicians, moves nothing—0.027, with a member-block bootstrap interval of $[-0.160, +0.228]$. Whatever δ measures, it is not an artifact of the switch variable, because the switch variable is switched on in that row too. Dropping the rung moves the pooled estimate by 0.004, so it is an exhibit rather than a

Table 6: Transfer rises with organizational distance

Break type	Breaks	Transfer
Re-badge: same organization, same site	5,457	0.027
Same organization, different site	7,424	0.270
Same system, different organization	47,141	0.507
Arm's length	78,109	0.552

Notes: Breaks are classified by how far the patient moved organizationally, from a second billing identity at the same address to an arm's-length change of owner. Transfer is the scaled jump defined in equation (4). The first row is an internal placebo: same physicians, same address, a different billing identity, and its member-block bootstrap interval is $[-0.160, +0.228]$, or $[-0.245, +0.487]$ clustered on the destination organization. The distance between the ends of the ladder is measured away from zero at both levels: arm's length minus re-badge is $[+0.320, +0.718]$ and $[+0.064, +0.800]$. The two middle rungs are reported as points because they were not bootstrapped.

filter.

A design built on movers can also be carried by a few procedure groups in which patients are unusually captive. It is not. A patient's return to her previous hospital divides into the part her new organization's own routing would deliver by chance and the part she brings herself, and the second exceeds chance in all ten procedure families—from 1.81 times chance in ear-nose-and-throat surgery to 2.92 times in neurosurgery and spine (Appendix E). Nor is the transfer a story about who reaches surgery at all: benchmarked against non-switchers on the same clock, a break raises the probability that a subsequent operation of the same family occurs by five to seven percentage points across elapsed-time bands—the same order as the intensive-margin cost of switching—so the estimates here are about where operations happen, not whether.

Table 7 closes the section by collecting its transfer coefficients beside the physician-level one of Section 3. They are grouped by how each is expressed, since a share of a gap and a slope on that gap are not comparable in level. The section's answer is the first row: about half of the difference between two organizations' hospital distributions travels with a patient who moves between them. It is the least processed number here, and the one the rest of the paper uses.

The second slope is the one not yet described: the transfer measured in *price* rather than in the identity of the destination, since organizations differ in how expensive their usual hospitals are. On the forced arm alone it is larger, 0.397 (0.079), which is the wrong direction for a story in which patients shop on price and the right one for a story in which the chain carries them. In economic units, moving from an organization at the tenth percentile of destination price to one at the ninetieth raises the price of a patient's operation by 19.9%, about \$454 at the median priced operation of \$2,279. Appendix D gives its specification.

What they share is a sign and the property that none of them is zero: a patient who

Table 7: Transfer of routing, by link and by estimator

Link	Object transferred	Estimate
As a share of the gap standing after the break		
Primary-care organization → hospital	destination distribution	0.497
Physician → new employer	the physician's own routing	0.64–0.68
As a regression slope on that gap		
Primary-care organization → hospital	destination distribution	0.300 (0.009)
Primary-care organization → hospital	destination price	0.247 (0.016)

Notes: The two panels are the same transfer computed two ways and are not comparable across the line, which is why one estimate appears in both, at 0.497 and 0.300. Both divide by the squared distance between the two organizations' distributions, and the candidate set is not what separates them; the baseline period is, being dated from the patient's first visit to the new organization for the level and the previous calendar year for the slope. On a common baseline the two agree. Within a panel the rows are measured on different samples and in different units, and are reported as a family rather than ranked—the smallest is not a weaker version of the largest but a different measurement. The destination slope is corrected for noise in the menus; the price slope and both levels are uncorrected and are therefore floors. Standard errors, in parentheses, are clustered on the destination organization. Neither row in the first panel carries one: the first is a bootstrap statistic whose interval is given with the arms it is compared against, and the second is a range across horizons rather than a single estimate. The rows are estimated, in order, in Table 4, Figure 1, and Appendix D for both slopes.

changes referring organization inherits a substantial fraction of the new organization's routing—which hospitals she reaches, and what they charge—without having chosen any of it.

The balance test lands where the design predicts. Within a break, a patient's own pre-break share at a destination and her new organization's leave-one-out share there are strongly related (+0.73), and the relation is nearly as strong on forced breaks whose roster moved to a single successor—where she exercised no choice of organization at all (+0.64). Read as a bound, sorting on tastes accounts for at most about a tenth of the correlation between what she wanted and what her new organization offers; the rest is the overlap of neighboring menus, which is the object of the paper rather than a threat to it.

5 What the Break Moves

A patient whose primary-care practice closes keeps her address, her insurance, her diagnosis and her tastes. What she loses is the set of hospitals that practice had been sending people to. Where her next operation happens is therefore informative about how much of a destination is carried by the relationship rather than by her, and the break is the only moment at which the two can be told apart.

The comparison that makes this concrete is the rate at which she returns to the hospital

she used before. A patient whose practice does not change returns there 43.1% of the time. One who changes to a practice that already sends patients to that hospital returns 38.4%. One who changes to a practice that has sent nobody there returns 15.9%. All three are measured on a matched clock, the non-switching arm reweighted to the switchers' distribution of elapsed time, because returns decay with time since the break and the arms are not otherwise observed on the same one.

The decomposition is the point. Changing organization at all costs about five points of return. Losing the hospital from the new organization's menu costs a further twenty-two. Whatever it is that binds a patient to a hospital, four fifths of it is not the act of changing practices; it is which practice she changed to.

Neither extreme survives this. A model in which the organization dictates the destination predicts that the off-menu rate is zero, and it is not: one patient in six still reaches a hospital her new practice never uses. A model in which the patient chooses predicts that all three rates are equal, and they are not: the gap between the second arm and the third is more than four times the gap between the first and the second. Both parties act, and the question is a quantity rather than a dichotomy.

The comparison generalizes, and generalizing it supplies its own control. Classify every hospital a patient used by whether her old organization sent other patients there and whether her new one does. Four cells result, and two of them are cells in which nothing about the menu changes: hospitals both organizations use, and hospitals neither uses. If the break moves destinations because it moves menus, those two must sit still while the other two move.

They do. The share of her operations at hospitals both organizations use is 85.3% before the break and 85.1% after; at hospitals neither uses, 2.2% and 2.3%. Against that stability the other two cells move sharply. Hospitals only the new organization uses take 5.1% of her operations before and 9.2% after. Hospitals only the old one used take 7.3% before and 3.5% after, a fall of just over a half. The four cells are measured on 136,008 pre-break and 141,447 post-break operations, and each set of four partitions her choices, so the movement between them is a reallocation rather than a change in how much surgery she has.

Two features of this deserve emphasis, because they are what make the design more than a correlation. The stationary cells are not assumed to be stationary; they are measured, and a patient whose tastes shifted at the break for reasons of her own would move them. And the cells are defined without reference to where she actually went—by what her two organizations do for their other patients—so the classification cannot be an artifact of her own behavior.

6 A Model of Inherited Menus

The facts describe a market in which someone assembles a short list and someone else picks from it. This section writes that down. The model has two purposes: to define the effective menu as an estimable object, and to make explicit which of its parameters the data can separate and which it cannot.

6.1 Two parties, one destination

Patient i requires an elective operation of procedure group g in year t. Let $\mathcal{J}_{igt}$ index the facilities that perform g within reach of her residence. Two parties bear on the outcome: her referring organization $r(i,t)$—a billing entity, not a physician and not a building—and the patient.

The organization's valuation of sending her to facility h is

$$V_{irh} \;=\; \kappa_{rht} \;+\; \nu_{irh}, \qquad \nu \sim \text{i.i.d. EV1}, \tag{5}$$

where κ_{rht} is the organization's **referral capital** at h—the stock of relationship, familiarity and routing habit that Section 3 shows is held by the organization rather than by its physicians—and ν_{irh} is an idiosyncratic shock to that valuation. The patient's utility is

$$U_{ih} \;=\; g(d_{ih}) \;+\; \beta_{\text{hosp}}\mathbf{1}\{h \text{ hospital}\} \;+\; \varepsilon_{ih}, \qquad \varepsilon \sim \text{i.i.d. EV1}, \tag{6}$$

in which g is quadratic in miles and carries an indicator for the nearest facility, so that the marginal disutility of distance is free to vary with how far she is already travelling. There is no facility effect in (6): one would absorb the hospital indicator, and Section 8 reports the facility-effects variant separately for that reason.

The organization offers h when V_{irh} exceeds the value of leaving it off the list, which carries a shock of its own; the difference of two extreme-value shocks is logistic, so h enters the patient's effective choice set with probability

$$\varphi_{irh} \;=\; \Lambda(\gamma_0 + \gamma' A_{irht})\,, \tag{7}$$

where Λ is the logistic function and A collects the observable correlates of κ: the leave-this-patient-out share of the referring organization's episodes at h, the analogous share for the specialist organization, an indicator for shared system ownership, and an indicator for whether h is a hospital. The specialist organization appears because severing that link moves the destination seven times as much as severing the primary-care link (Section 7). Both shares leave the focal patient out, so neither can predict her mechanically; in the inclusion equation they are the treatment, as advertising is in Goeree (2008), where in utility they would be an outcome and a bad control.

The share is an equilibrium object, and that is the standing threat to (7). Whatever makes a hospital attractive to an organization's patients in general—its size, its reputation, its case mix, how near it sits to the organization's catchment—raises both the share and the chance this patient is sent there, so the coefficient is not a causal weight unless that common cause is held. Leaving the patient out of her own share removes the mechanical part of the problem and none of this part. Two things address it: Section 8 identifies the menu coefficients from an organization's *deviation* from local practice rather than from a facility's popularity, which cuts the specialist-menu coefficient by more than half, and the transfer of Section 4 is estimated within patient and destination, where a facility's attractiveness differences out exactly.

Equation (7) is Goeree's inclusion rule: each facility enters or does not on its own, given *A*. Gaynor, Propper and Seiler (2016) reject that form for a competitive one in which a strong option displaces weak ones, and their objection applies here—an organization that takes on a destination plausibly drops another, so real menus are portfolios. What the competitive form buys is substitution inside the menu; what it costs is identification, and the estimate below reports what that cost turned out to be. Conditional on the realized menu C_i, the patient chooses $\arg\max_{h \in C_i} U_{ih}$, so that

$$\Pr(i \text{ chooses } h) \;=\; \frac{1}{1 - E_i} \sum_{C \ni h} \Pr(C)\, \frac{e^{U_{ih}}}{\sum_{k \in C} e^{U_{ik}}}, \qquad E_i = \prod_k (1 - \varphi_{ik}), \tag{8}$$

the normalization $1/(1 - E_i)$ conditioning on a non-empty menu.

The object of interest is the **effective menu**

$$M_i \;=\; \sum_{h \in \mathcal{J}_{igt}} \varphi_{irh}, \tag{9}$$

the expected number of facilities placed in front of her. Its sample mean is the paper's named parameter.

Distance is excluded from equation (7): the referring organization's inclusion rule contains no distance term, and distance enters only the patient's utility. This is the model's one substantive exclusion and it is what separates "she was never offered it" from "she did not want it." It is a restriction, and Section 7 treats it as falsifiable rather than maintained.

Price does not appear in equation (6), and this is a finding rather than an oversight, documented in Section 8: patient cost-sharing does not move the facility price a patient lands at, and elective surgery exhausts most patients' cost-sharing in any case. The model therefore sits in the zero-coinsurance limit familiar from Gowrisankaran, Nevo and Town (2015), where bargaining collapses to willingness-to-pay proportionality, and price enters the paper on the supply side rather than through patient demand.

Equation (5) makes the organization's capital the only state variable, which the facts do not support. Patients accumulate their own attachment: a patient's willingness to override an off-menu constraint rises with her own prior operations, and within a standing relationship a repeat operation of the same kind returns to the same facility 87% of the time.

The model accommodates this without a second equation, by adding her own prior use of h to (6) beside the organization's history in (7). Two inheritances then compete for the same destination: the organization's, which determines what she is shown, and her own, which determines what she picks from it. The rungs below hold the patient side fixed so that each is a statement about the menu alone, and Section 8 estimates the extension on its own: with her own-use terms in, her history takes roughly a third of the weight the specialist organization's menu carries without them, and neither term is a residual.

This structure has a boundary case worth naming. For the majority of operations in this sample the patient has no prior episode at all. There, her own stock is zero by construction and the destination is the organization's alone. That cell is the cleanest expression of the paper's mechanism and the one where the two-stock model reduces to the one-stock model.

Computing the likelihood, and two defects it hid

The likelihood (8) requires an expectation over $2^{|\mathcal{J}|}$ menus. I evaluate it by deterministic Gauss–Legendre quadrature on a bounded integrand rather than by simulation, which keeps the gradient smooth.

One thing the likelihood cannot do is worth stating, because the transfer of Section 4 does it and a reader may expect the model to follow. There the patient–destination fixed effect absorbs everything permanent about this patient at this facility, which is what makes a facility's attractiveness drop out. No such term is available here. A conditional logit normalizes within the choice situation, so an effect specific to a patient–*destination* pair is not absorbed by that normalization but has to be estimated, and on the mover panel there are of order four million such pairs. The consideration model is therefore estimated on movers rather than *within* them: restricting to patients observed on both sides of a break holds their address, insurance and illness fixed by construction, but nothing is differenced out. What that leaves is an assumption, and Section 7 states it and tests it against breaks the patient did not choose.

Two errors in earlier versions of this estimation are worth recording because neither was visible in fit statistics. First, the likelihood was initially *unnormalized*: without the $1/(1 - E_i)$ factor the model places positive probability on a patient considering nothing, and probabilities summed to $1 - E_i$ rather than to one. Second, the check that revealed it showed the original quadrature rule was itself inaccurate—probabilities summing to

0.879 at moderate node counts—which the change of variable repairs to eight decimal places. Both were found by asking a single question of the fitted model: do the choice probabilities sum to one? Neither appeared in the log-likelihood, the gradient check, or repeated-run determinism tests.

I report the Newton decrement rather than the optimiser's convergence flag. At these sample sizes the flag is unreachable: it tests an absolute gradient norm that does not shrink with n, and well-converged fits routinely return failure.

A specification that was tried and does not work

Gaynor, Propper and Seiler (2016) model the menu as a threshold in the referrer's own utility: h is offered if V_{irh} falls within τ of the referrer's most preferred option. That form is attractive here because τ has a direct reading as the degree of agency, and because inclusion is competitive—a strong option pushes weak ones out—which is the substitution structure a diversion exercise wants.

I estimated it and do not use it. The threshold is not differentiable, and smoothing it makes τ and the scale of V nearly collinear: the estimate ranged over three orders of magnitude across smoothing parameters, with the referrer coefficients moving in step. The likelihood is nearly flat along that ridge. A second problem is the sample: τ is identified by how often a patient reaches a facility her chain did not offer, which is an event that occurs at organizational breaks and not in a cross-section. Both are fixable with a hard-threshold estimator on the mover panel; neither is fixed here, and the alternative-specific form of equation (7) is what the results rest on.

7 Identification

The model has two sets of parameters that a cross-section cannot separate: the organization's capital κ, which determines what a patient is shown, and her preferences β, which determine what she picks. A patient who repeatedly uses one hospital is consistent with both. This section sets out the variation that separates them, states the assumption it requires, and names what the design cannot recover.

7.1 What the break moves

The identifying event is a **break**: a change in $r(i,t)$ caused by something other than hospitals. Practices close; physicians retire or leave the state; insurers reassign patients administratively. The identifying assumption is

$$\kappa_{r't} \;\perp\; \{U_{ih}\}_h \;\Big|\; \text{(exact ZIP, coverage, procedure group, calendar time)}, \qquad (10)$$

that is, *which organization she lands on is unrelated to her own hospital preferences*, conditional on the controls.

The entire logic is that a break shifts κ and leaves U alone. Her address, her insurance, her diagnosis and her tastes are the same on both sides; the party assembling her menu is not. This is the mover design of Finkelstein, Gentzkow and Williams (2016) applied to the referring organization rather than to a region.

The two parameter blocks are then identified from different variation. κ is identified from **menus**: the leave-this-patient-out distribution of an organization's other patients, which is estimable because organizations send many patients. The weight the menu carries is identified from the **off-menu return rate**. When her previous hospital h_0 is not on the new organization's menu, the only route by which she still reaches it is her own preference. The three return rates of Section 5 are the reduced form of exactly that contrast, and neither extreme model can produce them: a model in which the organization dictates predicts the off-menu rate is zero; a patient-only model predicts all three rates are equal. Formally, writing R for the event that her post-break episode returns to h_0 and $\mathcal{M}_B$ for the new organization's menu, the contrast that identifies the weight is

$$\Pr\big(R \mid h_0 \in \mathcal{M}_B\big) \;-\; \Pr\big(R \mid h_0 \notin \mathcal{M}_B\big), \tag{11}$$

evaluated within bands of elapsed time since the break, because returns decay and the two groups are not otherwise observed on the same clock.

Inside the fitted model the same contrast becomes a difference-in-differences on the inclusion probability at her pre-break hospital. Let $\bar{\varphi}_c^{\,\mathrm{pre}}$ and $\bar{\varphi}_c^{\,\mathrm{post}}$ be mean fitted $\varphi(h_0)$ before and after the break for arm $c \in \{\mathrm{on}, \mathrm{off}\}$. The estimand is

$$\Delta \;=\; \big(\bar{\varphi}_{\mathrm{on}}^{\,\mathrm{post}} - \bar{\varphi}_{\mathrm{on}}^{\,\mathrm{pre}}\big) \;-\; \big(\bar{\varphi}_{\mathrm{off}}^{\,\mathrm{post}} - \bar{\varphi}_{\mathrm{off}}^{\,\mathrm{pre}}\big). \tag{12}$$

The levels are not the object: h_0 is defined from her own pre-break choices, so $\varphi(h_0)$ is mechanically high before the break in *both* arms and only the difference of differences removes it. The placebo counterpart replaces the break with a fabricated one for a non-switching patient and computes the same Δ.

7.2 How each threat is met

The first threat is that she chose the new organization because of its hospitals. This is the first-order threat to (10) and it is met with the forced arm. Patients whose organization dissolved or whose physician left the market did not select a destination; those events are verified against federal provider records rather than inferred from missing claims. Transfer in the forced arm is statistically indistinguishable from the voluntary arm, with a confidence interval containing zero at both the member and destination-organization clustering levels.

Selection, where it appears, is visible in the *pre-period*: voluntary switchers drift toward their future organization's hospitals before the break; closures do not.

She may have moved house. Residential relocation would change both her organization and her distances. All specifications hold the exact ZIP code fixed, and the overlap between physician mobility and residential moves is 1.19%. The two margins are also orthogonal in what they move: comparing patients who relocate against patients who change organization, a house move changes how far her hospital is and not which one, and a break does the reverse.

The specialist may have re-chained. A break causes the specialist relationship to re-form, so conditioning on whether the specialist survived would be conditioning on a post-treatment variable. I do not condition on it: the estimand is the break's *total* effect, which includes the specialist channel.

The channel can nonetheless be decomposed, because each link can be severed on its own for reasons outside the patient. Classifying 468,707 consecutive same-procedure episode pairs by which links broke and why, the destination changes 15.2% of the time when neither link breaks. When the primary-care link is severed exogenously it rises by 3.7 points (1.8); when the *specialist* link is severed exogenously it rises by 26.2 points (1.1), a difference of 22.5 points (2.1). The specialist link's grip on the destination is roughly seven times the primary-care link's, and neither figure requires the sequential ignorability that a mediation analysis would. The timing agrees: after a primary-care break, the destination moves when the *specialist* re-chains rather than when the primary-care organization changes—among otherwise comparable post-break episodes, the patient leaves her previous hospital 58.6% of the time while her specialist is intact and 68.1% once he is not.

The links also turn over at very different speeds. Between consecutive same-procedure episodes the specialist link breaks 36% of the time, the facility link 21%, and the primary-care link 9%: the patient's durable attachments are her primary-care organization and her hospital, and the specialist is the fast link between them. That the fast link is the one whose severance moves the destination is not a coincidence but the mechanism, and one further contrast shows which end it is anchored to. When the specialist changes while the primary-care organization holds, 73.9% of patients are operated at the same facility as before, and while 41.0% end up at a facility the new specialist organization habitually uses, in three-quarters of those cases it is a facility the patient was already using—fewer than one patient in ten follows a new specialist somewhere genuinely new. The surgeon is replaced to fit the hospital rather than the hospital to fit the surgeon.

Two readings of that pair are available and the paper takes the weaker one. It could mean the primary-care organization matters little; but the two links are not independent, and breaking both moves the destination by more than the sum of breaking each (+0.108, s.e. 0.005). They behave as redundant anchors rather than as a chain in series, so the

primary-care link's small *marginal* effect is what a redundant anchor looks like when the other anchor is still holding, not evidence that it does no work.

Her own history is itself old referral capital. A patient's prior use of a hospital is itself the residue of some earlier organization's routing, so a model that lets her history predict her choice may be laundering referral capital as taste. The test is to estimate the menu where a carried history cannot exist. Restricting to patients whose operation is their first, preceded by at least a year of continuous eligibility with no earlier episode, and matching an experienced comparison arm on procedure group and year, the consideration coefficients are if anything larger among the naive: 35.6 and 13.4 on the specialist and primary-care menus for 157,852 naive patients, against 34.7 and 11.8 for 104,337 matched experienced ones. The menu's grip is fully present where no patient-side persistence is possible.

And menus are measured with error. An organization's menu is estimated from finitely many other patients, which attenuates. Every correction I can apply—restricting to decisive switches, requiring independent confirmation from the insurer's record, and using a leave-one-out habit—moves the estimate up. The reported figures are therefore lower bounds, and the direction is not a specification search.

7.3 Falsification tests

Three tests could have killed the design. The first test fabricates a treatment. Patients who never changed organization are assigned a fictitious break date and a fictitious destination drawn from the real distribution of destinations reachable from their own organization. The estimated jump is -0.091—opposite in sign, about a fifth of the real effect in magnitude, with no discontinuity at the fake date.

The second is internal to the data. Some apparent organizational changes are one firm relabeling itself: a new billing identifier at the same address under the same name. Referral capital should not transfer across a change that is only a change of name, and it does not: the jump is $+0.027$ with a confidence interval spanning zero, against $+0.55$ at arm's length.

The third looks for a population in which the mechanism should be absent. Nobody's primary-care organization routes an ambulance. The menu's contribution to model fit is 2.72 times larger for elective operations than for emergencies, and the effective menu is roughly twice as wide in the emergency arm. Two cautions travel with this test and are reported with it: raw coefficients are not comparable across the two arms because a logit identifies preferences only up to scale, and the elective and emergency arms differ in site of care, so the comparison is made within site.

A fourth test cannot discriminate, and is reported for that reason. Abaluck and Adams-Prassl (2021) identify limited consideration from violations of Slutsky symmetry, and the design transports here with the organization's habit in place of price. On 223,771

predetermined facility pairs the asymmetry has the predicted sign and is not distinguishable from zero ($+0.014$ $[-0.004, +0.032]$)—but a positive control shows the estimator cannot recover an asymmetry generated by the paper's own model, because the pairwise effect is diluted roughly elevenfold in a choice set of forty-eight. The null is about power, not consideration; Appendix F reports the derivation, both controls, and the power arithmetic, and the paper draws no verdict from the test.

A fifth does not cleanly pass, and I report it as such. Distance is excluded from consideration in (7), so the distance coefficient should be stable across institutional regimes that change the referrer's authority. In the shipped estimates the interaction is $+0.104$ with a standard error of 0.056, and in the weakest-authority stratum it excludes zero. The exclusion restriction is therefore supported by the emergency test and not by the regime test, and I do not claim otherwise.

7.4 What this design cannot do

The design has four limits, collected here rather than distributed through the results. It cannot decompose the composite objective into whose utility is being served. Only products of decision weights and preferences are identified; the within-menu index is not the patient's welfare and is not labeled as such anywhere in this paper.

It cannot recover the origin of a menu. Six years of claims cannot observe an organization's founding, so what put a hospital on a menu in the first place is outside the data.

It cannot separate the physician from the organization in the first three years of a physician's tenure, because assimilation takes that long.

And it cannot speak to which building a multi-campus organization used. The claim never records it, six address sources were tested, and none recovers it.

8 The Effective Menu

This section reports the estimates. It proceeds in the order the model was built: the two panels, the functional form of within-menu choice, the ladder, the effective menu itself, the battery that tests the fitted menu against a result it was not fitted to, and the two extensions that matter most for what follows—the patient's own history, and price.

8.1 Two panels, two purposes

The parameters are estimated on two samples, and the two answer different questions. The **cross-section** is a 3% random sample of members (the counterfactuals of Section 9 use a separate 5% draw): 64,682 members contributing 116,326 choice situations and

5.60 million episode–candidate rows. It is the frame on which the levels of the model are estimated, because it is representative of elective operations in the state.

The **mover panel** restricts to patients observed on both sides of a break in their referring organization: 84,375 movers, 308,706 episodes, 14.81 million rows. Exact residential ZIP is held fixed across the break by construction, so a mover's geography is the same before and after. This is the frame on which the mechanism is identified, because it is the only place where the referring organization changes while the patient does not.

The mover panel is not a random sample of breaks, and the direction of its selection should be stated rather than assumed away. Requiring an episode on both sides of the break keeps 84,569 of 2.38 million breaks that clear the volume and geography gates. Relative to the full spine, the kept breaks over-represent forced causes (closure 2.9% against 1.6%, physician exit 1.7% against 1.2%) and PPO coverage (55.1% against 42.7%), and heavily under-represent second and later breaks (9.4% against 33.0%). The over-representation of forced breaks runs in the paper's favor—they are the cleaner variation—but the PPO tilt does not, since PPO is the regime in which the referring organization has *less* authority. The mover estimates should therefore be read as conservative on that margin.

One further construction detail bears on interpretation: the patient's pre-break hospital is inside her post-break choice set in 81.7% of situations. The remaining 18.3% are the off-menu arrivals, and they carry the identification of how much the menu binds.

8.2 The form of within-menu choice

Earlier drafts of this project put distance in logs, which required a correction for the 7% of episodes at exactly zero measured miles—an artifact of ZIP-centroid geography rather than a true zero—and made every estimate conditional on the size of that correction. The benchmark literature does not log distance: Gowrisankaran, Nevo and Town (2015), Ho and Lee (2017) and Gaynor, Propper and Seiler (2016) all use levels with a quadratic, and the first two add an indicator for the nearest facility.

A specification grid on the cross-section settles it. Under facility fixed effects, the levels-plus-quadratic-plus-nearest form reaches a pseudo-R^2 of 0.3123 against 0.2702 for $\log(d + c)$—a gain of 4.2 points—and it removes the distance-floor machinery entirely, because nothing is undefined at zero. That form is used throughout what follows. The log form is retained as a frozen robustness row and is reported where the comparison is informative, but it is no longer the live specification.

Because a quadratic has no single coefficient, the object compared across models must be defined once and used everywhere. It is the **marginal disutility of distance at the median chosen distance**: with $U \supset \beta_1 d + \beta_2 d^2$,

$$m \;\equiv\; \left.\frac{\partial U}{\partial d}\right|_{d=\bar{d}} \;=\; \beta_1 + 2\beta_2 \bar{d}, \qquad R \;=\; \frac{m_{\mathrm{M0}}}{m_{\mathrm{consideration}}}, \tag{13}$$

both evaluated at the *same* $\bar{d}$ within a panel. Under the retired log form the same object is $\beta_d/\bar{d}$, so estimates under the two forms are comparable as ratios even though their coefficients are not.

Because the two fits in R share members, their sampling errors are correlated. I compute per-cluster influence functions for both fits, join them at the cluster, and apply the delta method to $\log R$ with $\bar{d}$ held fixed:

$$\widehat{\mathrm{Var}}(\log R) \;=\; \nabla'\Big(\textstyle\sum_c \psi_c\psi_c'\Big)\nabla, \qquad \psi_c = \big(\psi_c^0,\, \psi_c^1\big)', \tag{14}$$

with c the member (primary) and, in a second variant, the member $\times$ chosen-facility pair.

8.3 The model ladder

The ladder has four rungs.

- **M0**: the patient-only conditional logit. $\varphi \equiv 1$; every available facility is considered. This is the workhorse of merger review (Capps, Dranove and Satterthwaite, 2003; Gowrisankaran, Nevo and Town, 2015; Ho and Lee, 2017).
- **M1**: referral capital entered as a regressor in *utility* rather than in consideration. This rung exists to be rejected: it fits better than M0 but puts an endogenous quantity in preferences, and its coefficients cannot be read as tastes.
- **M2**: the consideration model above, estimated on a cross-section.
- **M3**: the same model estimated on within-patient variation—the same member observed on both sides of an organizational break, with exact ZIP code held fixed. Within-patient describes the frame, not the estimator: nothing is differenced out, and attachment to a particular building is measured by an explicit term rather than absorbed (the h_0 specification below).

Table 8 reports them, and Table 9 the maximum-likelihood estimates behind the consideration rungs. Each rung is estimated on the same frame within a column, so a ratio read down a column is a ratio; a comparison read across columns is not, because the two panels differ in composition.

A patient-only demand model estimated on these data overstates how much patients dislike distance by **88%** on the mover panel and 40% in the cross-section—before facilities are allowed intercepts of their own. With facility fixed effects the two overstatements fall to 17% and 9%: most of the gap between the patient-only and consideration rungs is facility heterogeneity the menu had been proxying, and the durable content of the correction is not a gentler distance taste but the account, reported below, of which trips the menu explains. The ratios in the table are exact for the specification the counterfactuals use; they are

Table 8: Marginal disutility of distance, utility units per mile at the median chosen distance

	Cross-section (median 8.07 mi)	Mover panel (median 8.91 mi)
M0 patient-only conditional logit	−0.1229	−0.1109
M1 menu entered as a utility shifter	−0.0887	−0.0704
M2/M3 menu as a consideration set	−0.0876	−0.0591
Overstatement, M0 ÷ consideration	1.40	**1.88**
95% CI, member-clustered	[1.35, 1.45]	[1.82, 1.94]
95% CI, member × facility	[1.12, 1.76]	[1.33, 2.65]

Notes: Marginal disutility of distance in utility units per mile, evaluated at each panel's own median chosen distance, as defined in equation (13). Each rung is estimated on the same frame within a column, so a ratio read down a column is a ratio; a comparison read across columns is not, because the two panels differ in composition. Confidence intervals come from per-cluster influence functions for both fits, joined at the cluster and combined by the delta method on the log ratio. Under facility fixed effects the ratio falls to 1.09 on the cross-section (same frame) and 1.17 on the mover panel (the forty-thousand-situation subsample whose no-effects ratio, 1.85, reproduces the full panel's 1.88); the text discusses what survives that absorption.

not robust to absorption, and the paper's quantities of record—the effective menu and the diversion shortfall—do not rest on them. The standard errors come from per-cluster influence functions for both fits, joined at the cluster and combined by the delta method on the log ratio: the two fits share members, so treating them as independent would err in an unknown direction. Two-way clustering on member and chosen facility widens the intervals substantially but does not bring either near one.

The two menu channels in Table 9 are not interchangeable, and their estimated weights reproduce, inside the model, the two-link anatomy that Section 7 established without it. The specialist organization's habit enters consideration with roughly three times the weight of the primary-care organization's (38.6 against 12.1 in the cross-section, 35.9 against 11.1 on movers; both regressors are leave-one-out shares, so the weights share units)—the same ordering the severance decomposition delivers model-free, where an exogenously severed specialist link moves the destination seven times as much as a severed primary-care link. The emergency placebo splits along the same seam: scaled against the distance coefficient, the specialist channel separates elective from emergency care by a factor of 3.3, the primary-care channel by 1.7. Model weights, severance arithmetic, and the falsification arm agree on the chain's structure. The primary-care organization and the hospital are the durable attachments; the specialist is the fast link between them; and the destination moves when that link re-forms, toward whichever hospital the receiving organization's habit names. The model's two coefficients are that sentence in likelihood form.

The intermediate rung deserves a careful statement, because the paper leans on it twice in opposite directions. As a structural object M1 is uninterpretable: entering the menu in utility makes it an endogenous regressor whose coefficient has no behavioral

Table 9: Consideration-set model: parameter estimates

	Cross-section		Mover panel	
	Coef.	(s.e.)	Coef.	(s.e.)
Consideration, equation (7)				
Constant	−4.567	(0.017)	−4.576	(0.012)
Specialist-organization menu	38.59	(0.83)	35.87	(0.55)
Primary-care organization menu	12.11	(0.26)	11.12	(0.14)
Referrer shares candidate's system	0.668	(0.027)	0.629	(0.017)
Hospital	0.182	(0.017)	0.090	(0.013)
Utility, equation (6)				
Distance (tens of miles)	−1.040	(0.023)	−0.730	(0.015)
Distance squared	0.102	(0.002)	0.078	(0.002)
Closest facility	0.249	(0.024)	0.301	(0.017)
Hospital	−0.190	(0.018)	−0.042	(0.013)
Candidate rows	5,599,200		14,808,971	
Members	64,682		84,371	

Notes: Maximum-likelihood estimates of the two-stage model of Section 6 under the benchmark distance form, with asymptotic standard errors from the inverse numerical Hessian. Menus are the referring organizations' leave-one-out destination shares; the likelihood is conditioned on a non-empty menu. Estimates for the patient-only and menu-in-utility rungs are not shown beside these: a logit identifies coefficients only up to the scale of its error, which differs across rungs, so cross-model comparisons are made in ratio form in Table 8. The mover-panel frame differs from the funnel's mover cascade (Table 1) by four members dropped in this frame's construction.

reading, and the paper never quotes its parameters. As a *reduced form*, however, M1 is the consideration model's twin, and the two are held to the same measured witnesses in Section 9.1: on realized displaced demand they perform alike, and both dominate the patient-only model by the same wide margin. The reallocation evidence therefore identifies the routing information's *content*, not its placement; the case for the consideration structure rests on what only it can express—the countable menu above, the revealed exclusions of Section 8.5, and the emergency placebo—and every screening conclusion in this paper is placement-robust by construction. The twin itself does not go unchecked: a moment-inequality estimator in the spirit of Ho and Pakes (2014), which differences out facility effects algebraically and imposes no choice form, bounds the habit's utility weight away from zero and contains the twin's fitted habit-to-distance ratio (-6.4 inside $[-23.3, -1.0]$). Where the twins finally separate is in time. Among 329,415 same-patient, same-procedure episode pairs inside one relationship, the patient repeats her facility 87% of the time. A utility model with fresh shocks each episode can produce at most its own concentration—50% here, however habit-loaded—while drawing the menu *once per relationship*, which is what this paper means by a menu, delivers 61% with no new parameters. Granting both architectures the building-level attachment the mover panel measures (+1.945, Section 4.2)

settles the rest: menus drawn per relationship then reproduce the moment—86% against the measured 87%, with nothing fit to it—while the taste model reaches 64%, and to rationalize the measurement must inflate the attachment to 2.6 times its measured value, roughly the disutility of seventy extra miles. Menus redrawn per episode saturate at 69% at *any* attachment strength, because a facility absent from the fresh draw cannot be loyal to. Snapshots make the two models twins; the same patient, revisited, tells them apart, and the separator is the mechanism itself: the menu stands.

8.4 The named parameter

The named parameter follows directly from the fitted inclusion probabilities. Averaged over episodes, a patient has **48.1** facilities within reach that perform her procedure, and

$$M \;=\; \sum_{h} \varphi_h \;=\; 2.39 \;\; [2.34,\, 2.44] \quad \text{facilities,} \tag{15}$$

where the interval is a 95% band combining a member bootstrap of the episode mean (standard error 0.004 over 111,567 members) with a delta-method parameter component (0.026) in quadrature; the parameter component dominates sampling seven to one, and even the worst-case bound on parameter correlations keeps the menu inside $[2.28, 2.49]$.

The mean inclusion probability across all candidates is 0.050; at the facility actually chosen it is 0.652. The gap between those two numbers is the model's entire content: the chain does not shade a wide field slightly, it puts two or three facilities in front of the patient and leaves the other forty-five effectively absent.

The level is robust across the specifications that could have moved it. Under the retired log form the effective menu is 2.26; applying the mover panel's stage-A parameters to the cross-section frame gives 2.24; the live form gives 2.39. The three differ by less than 7%, which matters because the level—not merely its ordering—is the number the paper reports.

Full consideration is rejected decisively rather than marginally. A model in which every available facility is considered is the $\varphi \equiv 1$ corner, which would place M at 48.1. Nothing in the estimates approaches it.

8.5 The menu, constructed from choices

The fitted menu is an estimate of an object nobody records. This subsection builds the same object a second way, from revealed preference alone, and asks whether the two constructions agree. Start from a practice's *potential* menu: every non-Kaiser facility offering the procedure group within fifty miles of the practice's patient centroid—a radius chosen so that it contains 91.8% of where the practice's patients actually go, not by convention. A typical practice-group-year faces 114 such facilities.

Membership is then settled from behavior, on both sides. A facility is *in* when at least two distinct members chose it and their count beats a noise process calibrated to the measured rate at which episodes escape the practice's routing (4.3%); it is *in at material share* when the lower confidence bound on its share clears five percent. A facility is *out at material share* when the upper bound falls below five percent—a one-sided claim with a stated error rate, powered by sustained zeros: a practice that performed hundreds of operations and sent none to a nearby offering hospital has said something statistically loud. Switching patients supply a sharper excluder: if her new practice's patients use her old hospital and she nonetheless moved to a new one, the new hospital was revealed preferred, so it cannot have been on the old practice's list—a construction that runs in both directions, polices itself (both memberships firing at once is a contradiction, and the rate is 8.3%, matching the independently measured 10.8% rate at which patients switch facilities with no menu change at all), and is refuted by direct choice evidence in only 4.3% of its organization-level claims. Pooling the six-year window where single years lack power classifies 92% of the potential menu.

The result is a countable list: the median practice uses about 14 of its 114 potential facilities for a given procedure group—about eight of fifty-eight hospitals and six of fifty-seven surgery centers, each tallied on the cells offering that site, so the site universes overlap the 114 rather than partition it—and the patient's effective menu of 2.4 is roughly one-seventh of it, a thinning the fitted model reproduces from entirely different inputs. The construction then enters the estimator as constraints rather than decoration: material membership puts a floor under the model's inclusion probability at that cell, and material exclusion caps the model's implied choice share. Audited at the fitted parameters, the floors are violated at 2.9% of thirty thousand certified cells against the certification's own five percent error budget, and the ceilings at 0.9%, with the model's mean choice share at excluded cells (0.006) sitting eight times below the cap. The model was never told about these bounds; it satisfies them at the rate a correct model would. This is also the content that separates the consideration structure from its utility twin: a taste can be small, but only a menu can be *off*, and only a menu model can be bounded by exclusions with error rates.

8.6 An out-of-sample test

The estimates so far are internal: they describe a likelihood that fits. The question that decides whether the menu is real is whether the fitted φ's reproduce something they were not fitted to.

They were fitted to choices. They were not fitted to *returns*—the rate at which a patient who has changed referring organization goes back to the hospital she used before. Section 4 measured that pattern without any model: after a break, a patient returns to

Table 10: Moments the likelihood was never asked to match

	Measured	Model
Chooses the nearest capable facility	0.207	0.197
Bypasses ten nearer facilities	0.455	0.466
Bypasses twenty-five nearer facilities	0.273	0.245
Stays inside the referrer's hospital system	0.309	0.299
Chooses a hospital rather than a surgery center	0.650	0.648
Returns to the pre-break hospital	0.367	0.353
Uses any pre-break facility	0.423	0.414
Mean distance travelled (miles)	20.3	18.5

Notes: Both columns are computed on the mover panel, so the comparison is model fit rather than a population match: the population share choosing the nearest capable facility is the 14.9% of Section 3, not the 0.207 here. None of these moments enters the likelihood. Model values come from the menu simulator carrying the non-empty-menu weights, whose choice probabilities sum to one on average across episodes. The two misses are stated in the text.

her old hospital 38.4% of the time when that hospital is on her new organization's menu and 15.9% of the time when it is not, against 43.1% for comparable non-switchers on a matched clock.

The model reproduces the shape. Fitted φ at the patient's pre-break hospital is **2.64 times** higher for on-menu than off-menu arrivals, and the difference-in-differences—the change in φ at that hospital from before the break to after, on-menu arrivals minus off-menu—is +0.2045. The levels are not the object to quote here: the pre-break hospital is defined from the patient's own pre-break choices, so φ at that hospital is mechanically high before the break in both arms. The difference-in-differences removes exactly that.

That is one moment. Table 10 reports eight, none of them targeted by the likelihood, measured and modeled on the same panel so that the comparison is fit rather than a population match. Six are within about a point: how often she is treated at the nearest capable facility, how often she bypasses ten nearer ones, whether she stays inside her referrer's hospital system, whether she goes to a hospital rather than a surgery center, whether she returns to her pre-break hospital, and whether she uses any facility from her own pre-break history. The closest is the choice between a hospital and a surgery center, which the model reproduces to within a sixth of a point without having been shown it.

Two are misses and are worth naming, because both are in the tail and both run the same way. The model sends patients 18.5 miles on average against a measured 20.3, and it understates the deepest bypasses—patients who pass twenty-five nearer facilities—by nearly three points. A consideration model built from an organization's habitual destinations will underpredict the rare long journey, because the habit contains few of them. Where the counterfactuals of Section 9 depend on the far tail, they inherit that.

The distance-band calibration behind the mean is closer than the mean itself: the

model puts 9.9% of episodes within two miles against a measured 10.3%, and 7.5% beyond fifty against 8.1%. The gap is concentrated in the outermost band, which is where the miles are.

Under the retired log form the same battery adds a third arm: a pseudo-mover placebo, in which a patient who does not switch is assigned a fabricated break date and run through the identical estimator. Her φ at h_0 drifts -0.082 over the fake event, against -0.122 for real on-menu arrivals and -0.323 for off-menu arrivals—1.5 and 4.0 times the placebo.

8.7 Two stocks inside the model

Section 6 argued that two histories bear on the destination: the organization's, and the patient's own. The estimates so far load all persistence onto the organization, which will make merger counterfactuals reallocate patients too easily. M5 adds the patient's leave-out pre-break use of each facility to the inclusion equation:

$$\varphi^{\mathrm{M5}}_{irh} = \Lambda\big(\gamma_0 + \gamma_1 h^{\mathrm{spec}}_{rh} + \gamma_2 h^{\mathrm{pcp}}_{rh} + \gamma_3 \mathrm{samesys}_{rh} + \gamma_4 \mathrm{hosp}_h + \gamma_5 \mathrm{own}_{ih} + \gamma_6 \mathbf{1}\{\mathrm{own}_{ih} > 0\}\big), \quad (16)$$

where own_{ih} is her pre-break share of episodes at h, computed leave-one-out within the pre-window so that her current choice cannot predict itself.

Her own history matters, and it takes some of the menu's weight: the specialist-menu coefficient falls from 35.87 to 24.44, roughly a third, while her own-use terms enter strongly (own share 1.354, s.e. 0.074; any prior use 3.350, s.e. 0.054). The distance coefficient falls in absolute value as well, which is the expected direction—some of what looked like an aversion to distant hospitals was attachment to a particular one.

The decomposition that follows is the reason for doing this. Simulating the break design inside the fitted model gives a jump of 0.334 under the full two-stock model and 0.415 when the patient's own stock is switched off, against a measured 0.403 on the same panel (below Section 4's event-based 0.497 because the panel and the baseline period differ; the comparison here is internal to one frame). A patient-only model is absent from this list because its simulated jump is zero by construction: nothing in it changes when the referring organization does. The organization's menu acting alone would slightly *over*-predict how far a patient moves when her referring organization changes; her own attachment holds her back; the measured value sits between the two. Neither stock alone reproduces the reduced form.

M6 adds three further controls that a referee will ask for:

$$\varphi^{\mathrm{M6}}_{irh} = \Lambda\big(\gamma' A_{irh} + \gamma_7 \log m_{a(r)h} + \gamma_8\, p_h \cdot \mathrm{managed}_i + \gamma_9 \mathrm{PPO}_i + \gamma_{10} \mathrm{exp}_i + \gamma_{11} \mathrm{sev}_i\big), \quad (17)$$

where $\gamma' A_{irh}$ collects the terms of (16). The new terms are the area's own facility distribution $m_{a(r)h}$, so that the menu coefficients are identified from an organization's

deviation from local practice rather than from facility popularity; a price term entering consideration on managed episodes only, with PPO as the within-model control group; and episode-level shifters of menu width. The specialist-menu coefficient falls further, to 18.82, as it must once common popularity is absorbed. The battery still passes—on/off band ratio 1.64, difference-in-differences +0.150—and M6 is the specification carried into the counterfactuals of Section 9.

8.8 Where price enters

Price is absent from patient utility in the main specification. Assuming price away and measuring it away are different acts, and this is the second.

Entering the patient's imputed out-of-pocket cost in dollars, interacted with coverage type, gives a coefficient of −0.005 for PPO members—who face real cost-sharing, 16.2 cents on the dollar—and −0.274 for members in managed products, who face 3.2 cents. The response is concentrated where the patient pays essentially nothing. Read as patient price sensitivity this is incoherent. Read as an institutional fact it is straightforward: managed products prune expensive facilities from the menu, and the price coefficient is picking up the pruning, not the patient.

That reading is testable, because the two stories differ within a person. Selection—patients who dislike expensive hospitals sorting into managed products—predicts the response follows the person. Pruning predicts it follows the current coverage. Among 25,904 members observed with episodes under both regimes, the facility-price coefficient is −0.0258 on their managed episodes and −0.0014 on their PPO episodes: the same people, and the response appears only where the chain has authority. As a bookend, the facility price *index* that an earlier draft used as a taste parameter is re-estimated on the same flip sample and remains unstable in the direction that disqualified it (−0.093 against +0.542); and under facility fixed effects the patient's own exposure enters *positively* (+0.075). A correctly signed response does appear once a facility is priced in expected out-of-pocket dollars rather than by an index—−0.031 pooled—but it is fifty times larger on managed episodes than on PPO ones, which is to say it is largest where the patient pays least. The paper reports that pattern rather than a single coefficient, and reads the response as the chain's rather than as hers.

This is a sign-and-pattern gate, estimated with flat-tolerance optimisation and without standard errors, and it should be read as such. Its value is that the two hypotheses predict opposite signs, so the test could have failed, and that a reduced-form design on a different sample reaches the same conclusion.

The reduced-form version. Compare the same person operating under a gatekept

and an open product—34,098 members observed under both:

$$\log p_{f(ie)} \;-\; \overline{\log p}_{C_{ie}} \;=\; \beta\,\mathrm{PPO}_{ie} \;+\; \mu_i \;+\; \lambda_{g(e)\times s(e)\times t(e)} \;+\; \varepsilon_{ie}, \tag{18}$$

the chosen facility's log price index against the mean over her own candidate set C_{ie}, with member and procedure-group-by-site-by-year fixed effects and standard errors clustered on the member. Losing the chain's authority moves her to dearer facilities—$\hat{\beta} = 0.116$ (0.004)—while her own share of the bill rises roughly fivefold, from 3.2% to 16.2%. The two hypotheses again predict opposite signs: if the patient were the price-sensitive party, removing the chain's authority should move her toward cheaper facilities, not dearer ones. The party that responds to price is the one the patient's plan design removes authority from. This is a within-member comparison and not a causal one—product is not randomly assigned—so I say "within-member" rather than "caused by".

8.9 Numerical discipline

Two defects in an earlier version of this likelihood were invisible in fit diagnostics and are described in Section 6. The checks that would have caught them are now run on every fit and reported here. Choice probabilities sum to one across the choice set to within 2×10^{-10}; the analytic gradient matches its numerical counterpart to a maximum relative difference of 5×10^{-8}; the quadrature rule integrates the menu distribution to 8×10^{-12} at 64 nodes.

The last of these needed a change of variable. The menu expectation in (8) is an integral over $[0, \infty)$ that Gauss–Laguerre handles poorly here: at 32 nodes the choice probabilities summed to as little as 0.879, at 128 the log-likelihood was still moving, and above roughly 192 nodes the weights underflow. Substituting $u = e^{-x}$,

$$\int_0^\infty f(x)\, e^{-x}\, dx \;=\; \int_0^1 f(-\log u)\, du, \tag{19}$$

gives a *bounded* integrand, and Legendre nodes on $[0, 1]$ with a graded map reach 8×10^{-12} at 64. Because $u^r = e^{-rx}$ exactly, only the nodes and log-weights changed—neither the likelihood nor its gradient.

Convergence is reported as the Newton decrement rather than the optimiser's success flag. BFGS uses an absolute gradient norm as its tolerance, which is unreachable at fifteen million rows: the cross-section consideration fit returns `success=False` at a Newton decrement of 1.5×10^{-7}, which is convergence by any decision-relevant standard. The decrement at every reported optimum is below 3×10^{-6}.

8.10 What the estimates do not settle

The stage-A coefficients are not structural parameters of a technology. They are the projection of referral capital onto four observable correlates, and their magnitudes move substantially with what else is in the equation—from 35.87 to 18.82 across M3, M5 and M6. What is stable across those specifications is the effective menu, the ordering of the bands, and the sign and rough size of the difference-in-differences. Those are the objects the paper carries forward; the individual γ's are not.

The distance ratio is a ratio of two fitted models on one panel, not a treatment effect. It says that a model omitting the menu attributes to distance what belongs to the chain. It does not say that the consideration model's distance coefficient is the true one—only that it is the one estimated with the routing structure in place.

And the model is estimated without facility fixed effects, so that the comparison across rungs isolates the specification rather than the absorption. That choice is not innocent, and the facility-effects version measures what it costs. On the cross-section, giving every facility its own utility intercept moves the consideration model's marginal disutility from -0.088 to -0.118 and the patient-only model's hardly at all, so the ratio between them falls from 1.40 to 1.09: most of the distance correction on that panel is facility heterogeneity that the menu was proxying. What survives absorption is the menu itself—the specialist-menu coefficient moves only from 38.6 to 35.3 and the effective menu rises from 2.4 to 3.2 of 48—so the correction to the *object* is robust where the correction to the *distance taste* is not. The mover panel, where the quoted ratio lives and where patient heterogeneity is already differenced out, gives the same verdict with the same machinery: the ratio falls from 1.85 to 1.17, the specialist-menu coefficient moves ten percent, and the effective menu is 3.1 of 48. Eighty percent of the distance correction, on both panels, was facility heterogeneity that the menu had been proxying. What the menu does once facilities carry their own effects is visible in where it improves fit: the gain is smallest for patients treated at the nearest capable facility and 2.7 times larger for patients who bypassed ten or more nearer ones, rising monotonically with the distance rank of the chosen facility. The menu's contribution is not a gentler distance taste; it is an account of the trips that distance cannot explain.

9 What Merger Review Misses

Merger review computes two things from a demand system: how much of a hospital's demand would divert to a rival if it became unavailable, and how much a plan's network would be worsened by losing it. Both come from a model in which patients choose. This section assembles the paper's punchline. It first measures where displaced demand actually goes, from two real shocks and with no model; shows that the standard model understates

within-system retention against that measurement at every radius, while the menu model tracks it; and traces what the two screens in use inherit from the error. It adds two market-level facts no screen observes at all: a hospital cannot buy its way onto a menu, and menus change hands by succession, which no filing captures. It closes with the ledger the patient inherits, on which money moves and measured health does not.

No real transaction is named anywhere in this section. The candidate pairs are hypothetical screening exercises constructed from co-occurrence in patients' choice sets; every flip rate is a property of a screen, not a prediction about any hospital.

9.1 Where displaced demand goes

When a hospital becomes unavailable, where does its demand go? Two shocks answer it before any model does, both now measured under the same ownership convention the models use and both carrying bootstrap confidence intervals. When referring organizations dissolve, the share of displaced demand retained inside the hospital's own system is 1.37 [1.30, 1.44] times what availability implies on the five nearest hospitals, rising to 2.84 [2.70, 2.99] on the fifty nearest—a member-block bootstrap over 3,132 closure breaks. When a hospital service line dies—the opposite side of the market, and an unrelated event—it is 2.51 [2.12, 2.99] times, on 363 events.

The same facility deaths also stage a forecasting contest between two rules built only from information available before the death. Asked where the displaced patients of a particular referring organization will go, that organization's existing habit beats proximity on every metric: mean absolute error 0.039 against 0.047, and it identifies the single largest destination 51.6% of the time against 23.4%. Rebuilt on the organization's *other* procedure groups—episodes disjoint from the ones being predicted, so the rule cannot be fitting its own outcome—the habit still wins.

The fitted models can be entered in the same contest, and this is the closest thing in these data to the stated second choices that discipline substitution in Grieco, Murry and Yurukoglu (2024): a hospital dies, its patients' demand must land somewhere, and where it lands is observed rather than surveyed. Across 594 organization $\times$ death cells and 777,293 realized post-death choices, each model is given the same displaced episodes with the dead facility struck from every candidate set, and its predicted distribution over the survivors is compared with the realized one. The patient-only model names the destination that actually absorbs the most displaced demand 22.0% of the time. Every model that carries the organization's routing information succeeds: the consideration model names it 49.1% of the time, the menu-in-utility variant 66.5%, and the organization's own pre-death menu 64.5%. None was fitted to these events. The internal ordering has a mechanical explanation—the utility variant loads the habit into choice almost directly, and the habit is the best single predictor of where displaced demand lands—while the consideration

Table 11: The closure experiment, measured and replayed inside the models

Choice set	Measured [95% CI]	M0 patient-only	M1 menu in utility	M3 menu
5 nearest hospitals	1.37 [1.30, 1.44]	0.57	1.11	1.21
10 nearest	1.54 [1.47, 1.63]	0.64	1.26	1.37
20 nearest	1.96 [1.87, 2.06]	0.81	1.60	1.74
25 nearest	2.16 [2.06, 2.27]	0.89	1.76	1.91
50 nearest	2.84 [2.70, 2.99]	1.17	2.31	2.51

Notes: Within-system retention relative to availability, on the 3,132 practice-closure breaks of the measured series: the same episodes, candidate sets, availability benchmark, and modal ownership rule for all three columns. Model columns replay each break with transported parameters (the cross-section patient-only fit and the mover-panel menu fit). Confidence intervals are a member-block bootstrap; because the comparison is paired, the gaps are estimated directly: at the fifty nearest, measured minus M0 is 1.67 [1.52, 1.81] and measured minus M3 is 0.33 [0.22, 0.44], and M0 lies below the measurement at all radii jointly in 100% of 2,000 draws. Parameter uncertainty and simulation noise are second-order (standard deviations 0.022 and below).

model's integration over latent menus costs it precision here, the same attenuation that appears everywhere else in this paper. The ordering that carries economic content is the other one: a twenty-two percent floor without the relationship, a forty-nine to sixty-seven percent range with it, under any placement.

The models, run on the same experiment. The sharpest comparison does not simulate a different counterfactual and reason by analogy; it replays the measured event inside each model. For every closure break above, each model is handed the same post-break episode and the same candidate set—which already carries the new organization's menus—and asked for the probability that the patient lands in her old facility's system, given that she changes facility. Measured and modeled retention are then computed on identical breaks, under the identical availability benchmark and the identical ownership rule, so the comparison supports a *paired* member-block bootstrap.

Among closure movers who changed facility, 43.8% stayed within the old facility's system. The patient-only model predicts 18.1%; the menu-in-utility variant, 35.7%; the consideration model, 38.8%. Because the availability benchmark cancels in any measured-to-model comparison, those numbers are the entire contrast: the patient-only model recovers 41% of measured within-system retention, and the two relationship-carrying models 81% and 88%, at every radius. Table 11 reports the ratios to chance.

Three readings, all disciplined by the pairing. The patient-only model is below the measurement at every radius, jointly in every bootstrap draw. The menu model's residual shortfall is also statistically distinguishable from zero: the correction closes about four fifths of the error and measurably not all of it, the pattern classical measurement error in the menus would produce. The obvious structural repair does not close it: letting a system's hospitals enter the menu *together*—a system-block copula on inclusion, sweeping

the within-system correlation from zero to 0.9 with every marginal preserved—moves the model's within-system diversion by less than a point and tops out below the measurement, so the residual sits in the inclusion margins themselves, not in their joint distribution; the leading candidate is attachment at the system rather than the building grain, which the replay's design removes for the closed practice's own hospital but not for its owner's sibling buildings. And the menu-in-utility variant lands beside the consideration model rather than beside the patient-only one—here and in the forecasting contest above. These measured events therefore identify *what* the models must carry, not *where* they must carry it: the referring organization's routing information is indispensable, and whether it restricts the choice set or shifts utility within a full one is not decided by any reallocation moment in this paper. The distinction that survives is the one only the consideration structure can express—a countable menu (Section 8.4), revealed exclusions with error rates (Section 8.5), and the emergency placebo—and the paper's screening conclusions require none of it: they follow from the routing information itself, under either placement.

The screening counterfactual. Merger screens do not remove referring organizations; they remove hospitals. Removing hospital j from every choice set in which it appears and re-solving the model gives the diversion matrix

$$D_{jk} \;=\; \frac{\sum_i \big[\Pr(k \mid \mathcal{J}_i \setminus j) - \Pr(k \mid \mathcal{J}_i)\big]}{\sum_i \Pr(j \mid \mathcal{J}_i)}, \qquad k \neq j, \tag{20}$$

the share of j's demand that reappears at k. The quantity to set against the shocks is how much of it stays inside j's own system, benchmarked on how much of the patient's choice set that system holds:

$$\rho_j \;=\; \frac{\sum_{k\in\sigma(j)} D_{jk} \big/ \sum_{k\in H} D_{jk}}{\mathbb{E}_i\big[\,(|\sigma(j)\cap\mathcal{J}_i|-1)\,/\,(|H\cap\mathcal{J}_i|-1)\,\big]}, \tag{21}$$

where $\sigma(j)$ is j's system and H the set of hospitals. A value of one is chance. Both the numerator and the denominator of (21) are shares among hospitals only: diversion also flows to ambulatory surgery centers and offices—roughly a third of it—and dividing a share of all destinations by a share of hospitals would deflate every model, the patient-only model most. Table 12 reports ρ under each model beside the measured series.

Three features matter, and the second carries the argument. **The ratio rises with the radius under every method, so no single number is the answer.** A system's hospitals cluster near its own patients, so widening the choice set dilutes the availability benchmark faster than it dilutes realized retention. In the measured series the numerator cannot move at all—it is where displaced patients actually went—so the entire rise from 1.37 to 2.84 is the benchmark thinning. Any comparison must therefore be read at a fixed radius, and the range must be quoted rather than a point.

Table 12: Within-system diversion relative to availability

Choice set	M0 patient-only	M3 menu	M6 full	Measured (referrer deaths)
5 nearest hospitals	0.95	1.38	1.38	1.37 [1.30, 1.44]
10 nearest	0.96	1.55	1.57	1.54 [1.47, 1.63]
25 nearest	1.09	1.88	1.95	2.16 [2.06, 2.27]
50 nearest	1.31	2.08	2.21	2.84 [2.70, 2.99]

Notes: Model columns simulate removing each hospital from the choice sets of the 5% cross-section; the measured column is the harmonized referrer-death series of Table 11, under the same modal ownership rule, with member-bootstrap 95% confidence intervals. M6 at the 50-nearest set is the specification carried into the screens below.

The hospital-removal exercise shows the same pattern as the replay. The patient-only model sits below the measured interval at every radius, and at the narrowest choice sets it puts retention *below* chance where the shock puts it half again above. The menu models sit inside the measured interval at five and ten hospitals and below it at twenty-five and fifty. The direction is the same everywhere and it is the direction that matters for policy: a model that understates within-system retention understates the harm from combining hospitals that already share an owner.

An earlier version of this comparison reported a sign error, and that claim does not survive. On a corrected benchmark the patient-only model is above chance at the widest radius, not below it. What survives is a systematic shortfall, which is a weaker claim and a better-founded one.

Two real shocks, on opposite sides of the market, put within-system retention well above chance. A patient-choice model estimated on the same data puts it lower than either, at every radius tested.

Two caveats remain, both smaller than they were. Ownership is now resolved by one rule—the modal system over a facility's panel years—in the measured series, the replay, the hospital-removal exercise, and the screens, so no part of this section compares conventions (Appendix G does, along with the replay's inference detail). And the replay removes the objection that measured and modeled quantities came from different experiments. What remains distinct is the facility-death witness, which removes a service line rather than a hospital, and the hospital-removal counterfactual itself, which corresponds to no event the data observe directly—that is what the replay is for.

9.2 Not a distance story

It would be natural to suppose that a model with referral menus simply makes patients less willing to travel, and that this is why more demand stays close to home inside a system. That is not what happens, and the distinction matters because distance is what

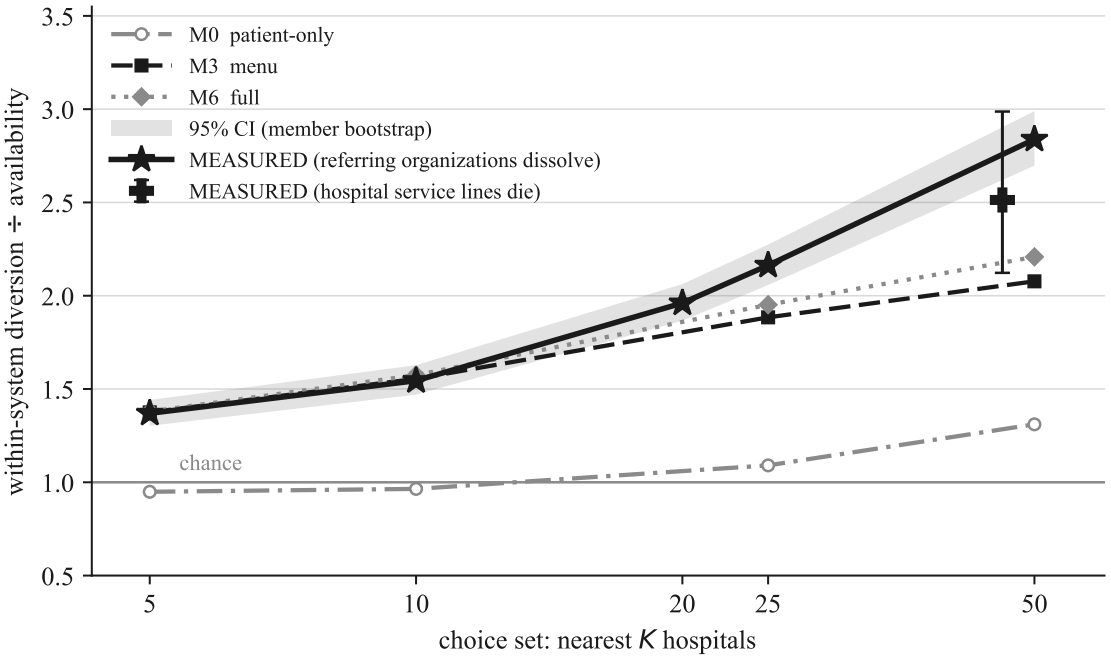


Figure 5: Two models and one measurement, at every radius

Notes: Within-system diversion divided by the share of the patient's available hospitals that the system holds; a value of one is chance. The starred series is measured from referring organizations dissolving, with its member-bootstrap 95% band; its numerator is fixed at where displaced patients actually went, so its entire slope is the benchmark thinning as the choice set widens. The plus marker is the independent facility-death measurement with its event bootstrap. All series use the same modal ownership rule. The model curves simulate hospital removal; the same-experiment comparison is Table 11.

the patient-only model already models well.

Figure 5 makes the point across radii. Both models send essentially the same share of displaced demand to the nearest alternative: 12.3% under the patient-only model, 10.7% under the menu model. Proximity is a null. What changes is *concentration*. The Herfindahl index of the diversion distribution is 0.037 under the patient-only model and 0.106 under the menu model, and the single largest recipient's share rises from 8.7% to 23.6%. Displaced demand does not travel less far; it travels to markedly fewer places, and to different ones.

The menu model also expresses something the patient-only model cannot. Under the menu model 30% of a closed hospital's demand goes to no hospital in the patient's consideration set—it is not recaptured by any rival, because no rival is on her menu. In a model where every available hospital is considered, this quantity is zero by construction. It is not a modeling artifact but the substantive content of a narrow menu, and it has no counterpart in the framework merger review currently uses.

9.3 What happens to the screens

Agencies use two screens, and they behave differently under the correction. **Diversion and GUPPI.** A diversion ratio is a probability under both models, so an absolute threshold means the same thing in each. The conventional test flags a pair when

$$\mathrm{GUPPI}_{jk} \;=\; \max\big(D_{jk},\, D_{kj}\big) \times \mu \;>\; \tau, \qquad \mu = 0.30,\ \tau = 0.05, \tag{22}$$

with the margin μ and threshold τ assumed rather than estimated, and both swept. Applied to a pool of pairs that are each other's leading substitutes, the patient-only model flags 0.5% of pairs. The menu model flags 5.0% and the full specification 7.0%: roughly ten to fourteen times as many. Among pairs that either model flags, the two disagree on 97.7%. The disagreement is nearly total, and it runs one way.

Willingness-to-pay. This screen requires more care, and an earlier version of this analysis got it wrong. The index is

$$\mathrm{WTP}_j \;=\; \sum_i \mathbb{E}_M\left[\ln \sum_{k\in M} e^{V_{ik}} \;-\; \ln \sum_{k\in M\setminus\{j\}} e^{V_{ik}}\right], \tag{23}$$

zero on menus not containing j, and a pair is flagged on its share w_{jk} of local-market index. A logit identifies V only up to the scale of its error. Because the index is nonlinear in that scale, two models with different error scales produce pair-share distributions in different locations for reasons that are not economic. Calibrating a cut on one model's distribution and applying it as an absolute threshold to the other then forces a flip rate of at least the difference in flag rates. At a top-decile cut that arithmetic floor is 32.2 points against an observed flip rate of 37.7: roughly 85% of an apparent "one in three verdicts flips" was the floor.

Cutting each model at its own quantile,

$$\mathrm{flip}(\alpha) \;=\; \Pr\Big[\mathbf{1}\{w^{\mathrm{M0}} > q_{1-\alpha}(w^{\mathrm{M0}})\} \neq \mathbf{1}\{w^{\mathrm{M}} > q_{1-\alpha}(w^{\mathrm{M}})\}\Big] \;\leq\; 2\alpha, \tag{24}$$

so that both flag the same fraction of pairs, is scale-free and closer to the problem an agency actually faces, which is a docket of fixed size. On that basis the screens disagree about 14.9% of pairs at a top-decile cut (the full specification; the menu-only rung flips 14.3%) and 8.9% at a top-twentieth—about one in seven, against a maximum possible disagreement of one in five. The residual is genuine, and it is precisely estimated: a member-block bootstrap of the screen—resampling the 89,571 patients whose episodes build the index, holding the candidate docket and the market definitions fixed—puts a standard error of under half a percentage point on the top-decile flip rate, with ninety-five percent intervals of $[0.143, 0.158]$ for the full model and $[0.137, 0.152]$ for the menu-only

rung. One in seven is not noise around zero, and it is also reliably below the one-in-five ceiling.

The most robust statement about this screen needs no cut at all. Ranking hospitals by their own index, the two models correlate at 0.905: they agree about which hospitals are important. Ranking *pairs* by their share of local-market index, they correlate at 0.443 ($[0.433, 0.451]$ under the same bootstrap). Consideration sets reshuffle local market structure even where they leave hospital importance nearly unchanged—which is precisely the level at which merger review operates.

What this implies for bargaining models. The screens are reduced forms of the negotiation models behind them (Gowrisankaran, Nevo and Town, 2015; Ho and Lee, 2017), in which leverage is what each side loses in disagreement, and both losses are computed from a patient-choice demand system: the enrollees' option value of keeping the hospital in network (Capps, Dranove and Satterthwaite, 2003), and the diversion the insurer expects when it steers around a dropped one. Under the measured menu both objects change shape rather than merely level. The option value concentrates: excluding a hospital costs nothing to the many patients whose menus never contained it, a great deal to the patients of the referring organizations whose menus did—and for 12.8% of episode-hospital pairs under the menu model (27.4% under the full model) the hospital is the menu's *only* member, so exclusion strands the admission rather than diverting it, the bargaining-table face of the thirty percent of displaced demand that reaches no hospital its patient would have considered. Leverage, in other words, is embeddedness in referral menus—a stock of relationships—which is also why acquiring referring organizations is a bargaining move and not only a routing one (Cooper et al., 2025). Recapture between hospitals that share menus is understated in the replay's proportion, so simulated price effects of same-market system mergers are understated in the same direction the screens err. One assumption comes out strengthened: the static protocol treats the diversion matrix as fixed over the contract horizon, and the menus' measured rigidity (Section 3) is exactly that.

The steering instrument is the sharper point. In Gowrisankaran, Nevo and Town (2015) the insurer steers patients through their own coinsurance-scaled out-of-pocket price—the mechanism shuts down at zero coinsurance precisely because the insurer then has no price with which to move the patient—and the model's central policy counterfactual restrains hospital prices by raising patient coinsurance tenfold. In this setting the steering is measured directly, and it does not run through the patient: tripling her cost-sharing exposure changes where she goes but not toward cheaper (Section 10.1), while shifting routing authority to the chain—within the same member—does move her down her choice set's price distribution (Section 8). The steering the bargaining model needs exists, but its transmission is the chain's authority and the menu's composition rather than the patient's coinsurance elasticity. A patient-side price coefficient estimated here loads on menu

composition rather than preference—the pruning, not the patient—and a counterfactual that moves patient cost-sharing moves the one agent in the chain who does not hold the pen.

A screen result that does not survive scrutiny. Under the menu model, 60.5% of large hospitals have a different closest competitor than under the patient-only model. That figure overstates its own content: 51% of hospitals have less than a ten-point gap between their first and second closest competitors, and two *consideration* models disagree with each other on 25.4% of the same hospitals. Much of it is the fragility of an argmax over near-ties. I report the number and do not build on it.

9.4 A hospital cannot buy onto a menu

The screens above price the demand a merger transfers. How much that matters depends on whether the same demand could instead be won in the market: can a facility *acquire* referrals it does not already have? If entry can contest a menu, a screening error is self-correcting; if it cannot, the demand a merger moves is demand competition will not reallocate.

A clean test is available at the entry margin. When a facility begins offering a procedure group it has never offered before, the organizations that might send it patients are not chosen by the facility, and any pre-existing relationship was formed for unrelated reasons and cannot have been caused by the new service. There are 282 such launches in 2019–2022, defined as a (facility, procedure group) pair appearing for the first time with at least twenty operations. Most are ambulatory surgery centers rather than hospitals, and the procedures are the ones that migrated to outpatient settings over this period—endoscopy, ophthalmology, dermatology—so the result should be read as being about outpatient entry.

Across all 282 launches, organizations that already send the facility work in *other* procedure groups adopt the new line at 38.2%; organizations that never use it adopt at 4.9%. Estimating adoption on the pre-existing relationship and distance jointly, with launch fixed effects so the comparison is across organizations facing the same new option and errors clustered on the organization, a prior relationship raises adoption by **0.257** (0.020). A facility that opens a new service line wins the referrers it already had and almost no others: relationships do not merely hold demand in place, they determine who can compete for it.

That estimate uses the cleanest available definition of a launch, and the reason it is not the obvious one matters. Reading the launches by facility name shows an academic medical center apparently adding thirteen procedure groups in a single year, and another twenty-four. Hospitals do not do that: the facility *arrived* in the claims data—a payer contract, a change of billing identity—and every group it performs registered as a first appearance. Because those facilities were billing beforehand at low volume, no pre-period

test detects them. 137 of the 282 launches come from sixteen such facilities. Restricting to facilities that add a *single* procedure group in a year removes those, and the smaller multi-group cases with them, leaving 91 launches; the estimate above is computed on those. Requiring only that a facility add at most three groups is looser and leaves 144, on which the estimate is 0.248 (0.021).

Three further checks bound the rest of the measurement error, and none of them moves the answer. Organizations that already use a facility are closer to it—fourteen miles against twenty-seven—so the relationship might be residual proximity; replacing the single distance term with twenty distance bins moves the coefficient by 0.001. The facility-service panel records nothing below three cases a year, so a launch date can post-date a facility's first case; verified against the raw records, 195 of 282 launches have zero prior episodes and 85% of them are still performing the procedure two years later. And risk-set membership uses volume in the same window as the outcome; defining it on pre-launch volume gives 0.279 (0.034). Across all seven definitions the estimate lies between 0.196 and 0.279, always more than six standard errors from zero; Appendix C tabulates all seven.

In distance units the relationship is worth between 3.3 and 5.6 log-miles depending on the definition, against a risk set spanning 4.6 from end to end—so within the observed range proximity cannot substitute for it. I report the adoption effect rather than a mileage because the conversion saturates: at these magnitudes almost any estimate maps to roughly the median distance, which makes a mileage look more precise than it is.

9.5 Demand moves at succession

If no living agent chooses the menu, demand should reallocate only when the lineage holding it breaks.

Consider the 70,674 patients who have the same type of operation in 2018–19 and again in 2022–23, classified by what happened to their chain in between. The four-year separation is chosen for exposure rather than for sample: succession is a slow process, and narrower windows would roughly double the number of patients while leaving less time for an organization to die. Holding the procedure group fixed matters too—a patient whose 2018 colonoscopy is followed by a 2022 knee replacement changes hospital for reasons that have nothing to do with her chain.

Figure 6 shows the result, and the ordering is the fact. Every broken link raises the chance a patient ends up somewhere new, and it rises with how central the broken link is: a specialist change moves fewer patients than a primary-care change, and the destruction of the facility itself moves almost all of them.

Aggregating over the intervening events, at least 36% of same-service hospital switching is attributable to succession rather than to drift—a lower bound, because even patients classified as drifting land on their current organization's leading hospital far more often

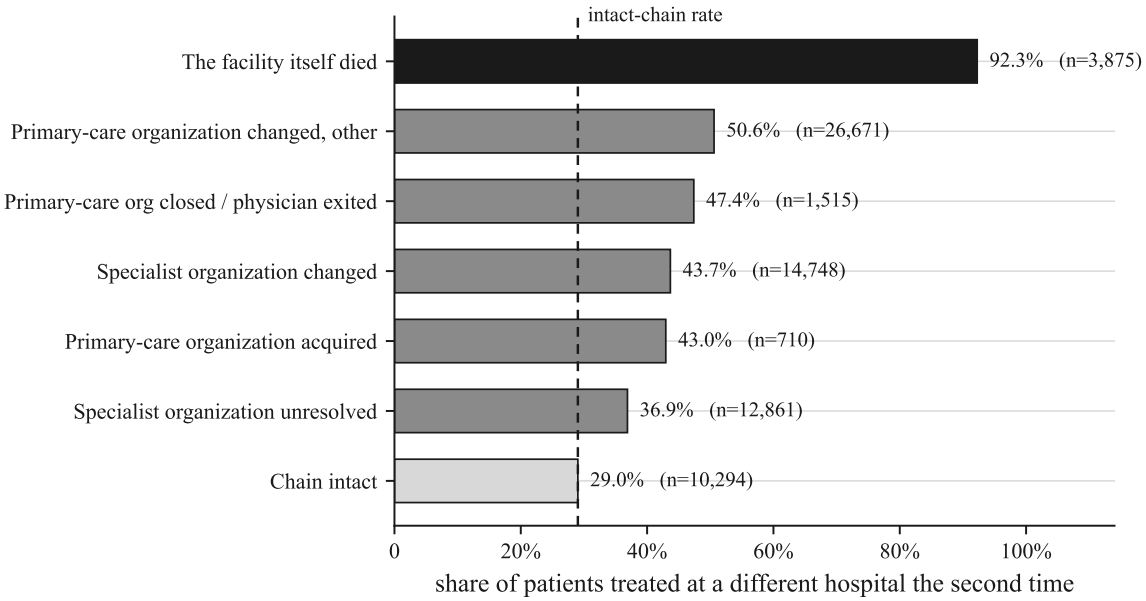


Figure 6: Hospital switching, by which link of the chain broke

Notes: Patients with the same type of operation in 2018–19 and again in 2022–23, classified by what happened to their referral chain in between, mutually exclusively and in priority order. Bars give the share treated at a different hospital the second time, with patient counts beside each. The dashed line is the intact-chain rate. The top bar is close to mechanical—a patient cannot return to a hospital that has stopped performing her operation—and is the ceiling the others are read against. Over four years 27.6% of these patients also change insurer, and those who do switch hospital more often (53.5% against 43.0%). Restricting to patients whose insurer is the same in both windows leaves every bar one to four points lower, the ordering unchanged, and the share attributable to succession at 35.6% against 36.4%: the insurer adds a roughly uniform amount of switching and does not generate the gradient.

than proximity predicts.

The reallocation is not chosen: patients whose organization dissolved or whose physician left the market transfer their routing at the same rate as voluntary switchers, as Section 4 established.

An intact chain is the exception. Only 14.6% of repeat patients kept both their primary-care and specialist relationships intact across four years. The machinery is coming apart continuously; hospital shares over the same period correlate at 0.979. Persistence at the market level is produced by replacement, not by stability underneath.

9.6 Concentration in the layer nobody screens

Succession does not only reallocate patients; it concentrates the layer doing the allocating. If demand is allocated by referring organizations, then concentration among *referrers* is an allocative fact about a hospital market, whether or not any hospital merges. Table 13 and Figure 7 measure both layers on the same geographic markets with the same weights,

Table 13: Concentration in two layers of the same markets

	2018	2023	Change
Referring-organization HHI	608	766	+25.9%
Hospital-system HHI	1812	1884	+4.0%
Top-10 referrer share	0.518	0.568	+5.0 pts

Notes: Herfindahl indices on the same three-digit ZIP markets with the same weights: referring organizations by share of episodes routed, hospital systems by share of episodes received. Group-practice ownership vintages are not used in this series.

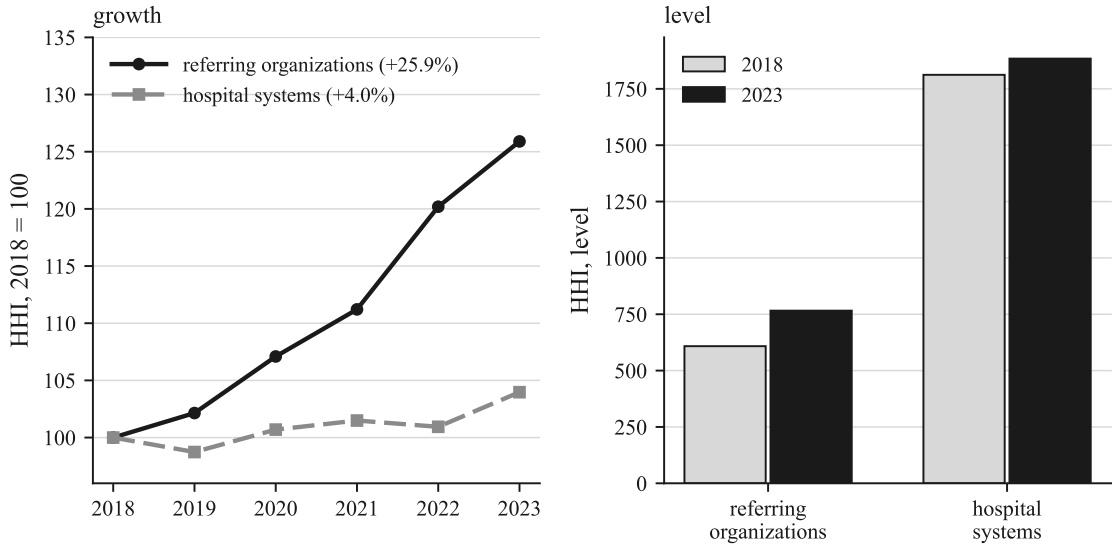


Figure 7: Two layers of the same markets

Notes: Herfindahl indices computed on the same three-digit ZIP markets with the same weights: referring organizations by share of episodes routed, hospital systems by share of episodes received. The left panel indexes both to 2018 to show growth; the right gives levels, which is the qualification the growth panel alone would hide—the referral layer is consolidating far faster from a much lower base. Group-practice ownership vintages are not used in this series.

over 2018–2023.

The referral layer is far less concentrated in level and is concentrating about six and a half times faster, monotonically across the six years. The hospital layer—the one merger review watches—is close to flat.

Two qualifications keep this in proportion. The referral layer starts from a much lower base, so equal proportional growth is a smaller absolute change; and an HHI computed on referring organizations is not a market-power statistic in the sense a hospital HHI is, because referrers do not set hospital prices. What the comparison establishes is narrower and still consequential: the layer that determines which hospital a patient reaches is consolidating quickly, while the layer that is monitored is not.

And it consolidates by the mechanism just documented: demand moves between

referring organizations through dissolution and absorption rather than acquisition. There is no transaction to review, no filing threshold to clear, and no notification requirement. The allocation of hospital demand is being rearranged by a process that antitrust has no instrument pointed at.

9.7 What inherited menus cost

The screens above price what routing error does to hospitals. The menu also writes a ledger for the patient who inherits it, and the paper can close that ledger on two margins: what riding an inherited menu does to the price of her operation, and what severing a chain does to her health. One entry is large and the other is bounded near zero, and the contrast is the finding.

The price entry is the transfer slope of Section 4 expressed in dollars. Organizations differ widely in how expensive their usual destinations are—the ninetieth-to-tenth percentile ratio of the facility price index is 11.9 across the 78% of frame episodes that carry a price (the remainder is capitated and structurally unpriced, counted rather than imputed)—and a quarter of that difference travels with a patient who changes organizations: 0.247 (0.016), uncorrected for noise in the organization-level price and therefore a floor. Moving from an organization at the tenth percentile of destination price to one at the ninetieth raises the price of the same operation by 19.9%, about $454 at the median priced operation of $2,279. On the forced arm alone the slope is larger, 0.397 (0.079), which is the wrong direction for patients shopping their way out of expensive inheritances and the right one for a chain that carries its prices with it. The pass-through is not symmetric—0.37 toward cheaper destinations against 0.21 toward dearer ones—an asymmetry in the direction a residual patient brake on price increases would produce, and the one place in this ledger where her own hand is visible. Who holds prices down is the same story run within a person: when a member's product shifts authority from the chain to her, her operations land 11.6 log points dearer against her own candidate set while her share of the bill rises fivefold (equation 18). The price discipline in this market belongs to the chain, so the price of an operation is, to a measurable degree, an inherited attribute.

The health entry is bounded near zero. If menus encoded patient-specific match—the agency story in which the chain knows which hospital is right for her—then severing a chain should hurt. The paper measures the severed: patients whose organization dissolved or whose physician left the market, compared with non-switchers assigned pseudo-breaks on the same clock and in the same cells, with enrollment required on both sides of the window. Thirty-day readmission after surgery is no higher for them: the point estimate is 2.13 percentage points *lower* (1.16), and the upper end of its ninety-five percent interval caps harm at 0.15 percentage points on a base of 5.4%. Emergency-department use within thirty days of the break, an access margin rather than a quality one, moves 0.4 percentage points

(0.4), indistinguishable from zero. Two clauses bound the claim. The forced inpatient cell is small—248 operations—so the interval, not the point estimate, is the result; and voluntary switchers are not quoted here at all, because patients who choose to leave differ from those who stay in ways no matching repairs. Thirty-day surgical readmission is not everything care could protect; it is the margin the data can see.

Money moves and measured health does not, and the two facts together are what license the reading of Section 10.3: the menu behaves like a claim on a payment stream, not like a store of match information. For merger analysis the asymmetry is the point. The margin on which inherited routing is consequential—which system is paid, and how much—is exactly the transfer margin the screens mismeasure, while the margin on which it is inert—patient outcomes—is the one a defense of incumbency would have to invoke.

9.8 What follows, and what does not

This is a measurement claim and should not be inflated into a policy claim it cannot carry. What follows is that a demand system omitting the referral menu understates within-system retention, and does so in the direction that clears combinations rather than flags them; that the two screens in use disagree substantially about which pairs to examine, with the disagreement concentrated in local market structure rather than in hospital importance; and that a consequential form of concentration in this market is invisible to both screens because it is not produced by transactions.

What does not follow is any verdict about a particular merger. The pairs here are hypothetical, the margin and threshold in the GUPPI screen are assumed rather than estimated, and the within-menu index is not a welfare measure and is not converted to dollars anywhere in this paper. The dollar magnitudes the paper does support are patient-level and sit upstream of the screens: a p10-to-p90 change of referring organization moves the price of the same operation by 19.9%—about $454 at the median—and removing the chain's authority moves a patient to facilities 11.6 log points dearer relative to her own choice set (Section 9.7). The routing the screens mismeasure is the routing those dollars ride on. Nor does it follow that the menu model's numbers are the correct ones: what the external shocks establish is that it is closer, at every radius, than the model in use.

Whether menus can be held fixed. Every counterfactual in this section holds menus fixed, and the paper's own evidence says where that choice is safe. For the removal class it is validated directly: the replay holds menus fixed through real organizational deaths and tracks realized diversion. For the merger class it is supported by measured rigidity: menus barely drift year to year against the differences between organizations (Section 3), and a facility cannot buy its way onto one at the entry margin (Section 9.4). The margin that could rewrite menus is ownership of the referring organizations themselves, the channel Cooper et al. (2025) document in Medicare. The paper does not price that

channel, but its direction is not neutral: to the extent employing referrers pulls their routing toward the acquirer, a merged system that also acquires practices retains *more* than a fixed-menu counterfactual predicts, and the understatement documented here is a floor. The error from holding menus fixed runs in the same direction as the error from omitting them.

10 What Inherited Demand Means

The paper to this point has treated inherited demand as a measurement problem: who routes demand, and what mismodeling the router does to the quantities merger review computes. This section asks what kind of arrangement the measurement has uncovered—for the patient inside it (Section 10.1), for the firms that hold it (Section 10.2), for what the asset contains (Section 10.3), and for the class of markets that share its structure (Section 10.4).

10.1 Nobody is wrong not to choose

Start with the patient, because the temptation is to read this paper as a story about her passivity. The data say something more specific: she behaves the way a rational principal with episodic stakes, no price signal, and a free agent should.

Delegation costs her nothing the data can price. Severing her chain does not harm her surgical outcomes (the ninety-five percent bound on harm is 0.15 percentage points of readmission on a base of 5.4%; Section 9.7), and the chain's authority is what gets her prices down (Section 8), while she pays 3.2 cents on the dollar rather than 16.2. Nor is she absent from the choice: her override of the menu is proportional to her own stock. Among switchers whose previous hospital has fallen off the new organization's menu, the share who return to it anyway rises from 12.1% with one prior operation to 16.7% with two and 28.3% with three or more. She intervenes exactly where she holds referral capital of her own, and delegates where she holds none.

The sharpest cells are the ones where the product hands her the pen anyway. Under a PPO her override roughly doubles (19.7% against 9.3% under an HMO)—and the same shift of authority, compared within the same member, moves her to facilities 11.6 log points *dearer* relative to her own choice set (Section 8). Exposure does not buy shopping either. Across terciles of cost-sharing exposure, 6.7% of the bill against 18.7%, her destination distribution genuinely changes: the total-variation distance between the terciles' distributions, the share of episodes that would have to move to make them identical, is 0.39 against a 0.25 floor from splitting the same organization's patients in half. But the price position of where she lands is flat: a price index of 1.19, 1.19, 1.18 across the terciles. Stakes without information move her around; they do not move her toward cheaper. The patient is not a suppressed chooser waiting to be liberated. On the one margin where the product liberates her, autonomy without information makes her prices worse.

The aggregate consequence is not that the demand response merger review counts on has vanished; it is that it lives somewhere else and has a different shape. What disciplines a hospital in this market is the chain's price sensitivity when the payer arms it, and the payer's decision about where to put that authority, not patient substitution across forty-eight candidates. And the arrangement enforces itself. Because demand routes through

chains, hospitals compete for menus, which entry cannot contest and only ownership moves (Section 9.4); because hospitals do not compete for patients, no individual patient gains anything by shopping; so delegating remains rational for each patient in every period. Nobody chooses is not an anomaly of this market but an equilibrium of it, and Section 9 is that equilibrium's measurement corollary: model it as patient choice and the error is not noise around zero but the signed understatement documented there.

This also inverts where the friction sits relative to the literature on inertia in health-plan choice (Handel and Ho, 2021), which locates the stickiness inside the consumer: inattention, switching costs, defaults she fails to leave. The stickiness here is outside her. The frozen state variable is the organization's menu, which she never held: a population of fully attentive, fully rational patients would produce the same frozen allocation, because the allocation was never theirs. Arrow's (1963) answer to uncertainty in medicine was the physician-agent, an agent valuable because she carries knowledge; what the severance evidence shows this agent carrying is a habit. The friction is institutional rather than psychological—which is why remedies addressed to psychology, information delivered to the patient at the moment of choosing, have so little surface to act on. If decision support has a moment, it is not the operation but the attachment years upstream, the one choice in the sequence that no one experiences as choosing a hospital.

10.2 What a hospital buys

If the patient does not hold the choice, someone's balance sheet does. The paper's results, collected in one place, describe an intangible asset with measurable properties. It is durable: a menu's year-to-year drift is an order of magnitude smaller than the differences between organizations, and the menu survives the replacement of the physicians inside it (Section 3). It cannot be contested from outside: a facility that opens a new service line wins the referrers it already had (Section 9.4), and a same-street rival's habit predicts nothing that proximity does not (Section 3). And it pays its owner: when a practice dies, the demand it routed stays within the owner's hospital system at up to nearly three times availability (Section 9.1).

Ownership is not incidental to that last property; it is what makes the asset heritable. When a practice held by a hospital system closes, its patients stay with the owner's hospitals at 15.7 percentage points above availability ($[15.2, 16.2]$). When an *independent* practice closes, the excess is -0.5 points ($[-1.7, +0.6]$): its demand scatters at exactly chance, and no excess flows to other independents either.[1] An independent organization's referral capital is an annuity that stops at its death. A system-held practice's is bequeathable. Acquisition, on this reading, does more than redirect flows: it converts mortal capital into

[1] Computed under the year-by-year ownership convention; Appendix G reports the measured series under both conventions.

heritable capital.

What the asset transmits is ownership, not merit. When a house move displaces a patient from her hospital, the replacement shares the old hospital's *owner* one and a half times as often as the distance-matched alternatives it was chosen over (1.50×); its CMS star rating is no closer to the old hospital's than a random alternative's (1.00), and its readmission rate barely closer (0.95). Diversion measured from a real displacement rewards holding the channel, not being better at the end of it: a hospital cannot earn this demand by improving, only by owning the organization that routes it.

The asset can also be measured against the one rival claimant to it: the patient's own physician. Take the breaks where a practice dissolved, and split them by whether her own physician resurfaces at the destination organization. Where the physician does, with the relationship intact and only the firm changed, she still transfers 0.375 of the organizational gap; where the physician does not, 0.492. The personal relationship accounts for roughly a quarter of the transfer and the organization's menu for three-quarters, and the voluntary arm shows the same split, with the difference carrying confidence intervals that exclude zero under both clustering levels. This is measured in California, where non-competition covenants have been void by statute for over a century, so it describes the free-mobility world directly: physicians can leave, do leave, and take about a quarter of the routing with them. In the debate over physician non-competes, the employer's claim that the covenant protects client relationships is, on the routing margin, a claim about that quarter.

That is what a hospital buys when it buys a practice. Not the physicians, who are two-thirds re-routed within three years of changing employer in whichever direction the new organization points (Section 3), but the menu: a claim on future admissions with near-zero measured drift. The physician-practice acquisition wave is, on the demand side, a market for exactly this asset, and it is a market review rarely sees. Hospital mergers transfer capacity and are examined; practice transactions transfer demand and are typically too small to trigger premerger notification (Wollmann, 2019; Cooper et al., 2025), while succession transfers the same asset with no transaction to file at all (Section 9.6). Antitrust watches the market for beds; the market for admissions clears elsewhere. What this paper establishes is the asset side: existence, durability, heritability. The routing consequences of purchase are documented in Medicare by Cooper et al. (2025); the direct test available here is underpowered, and Section 9 flags rather than prices that margin.

10.3 Rent, not information

An asset this durable should contain something. The natural candidate is knowledge: the referral is the transaction of a knowledge hierarchy (Garicano, 2000), and Arrow's agent earns her role by knowing what the patient cannot. That theory makes three predictions here, and the measured margins run against all three. If the menu encoded patient-specific

matching, severing it should harm the patients severed; the bound of Section 9.7 says it does not. If the knowledge were clinical, the menu should be procedure-specific; instead the same organization's habit predicts diversion even for procedure groups disjoint from the ones it was measured on (Section 9.1). And if the menu tracked merit, transmission should follow quality; it follows ownership (Section 10.2).

The reading the evidence favors is different: the menu is relational capital, in the sense of Baker, Gibbons and Murphy (2002). A stable referral flow is the currency that sustains cooperation between organizations, buying access for the sender's patients, scheduling priority and information flowing back, so a referral is less a diagnosis than an installment payment on a relationship. That is why the flow attaches to the continuing organization rather than to its people or its label: menus survive the replacement of the physicians inside them (Section 3), and a change of billing identity alone moves nothing (Section 4). It is why the relationship anchors at the institution rather than the individual counterpart, quantifying Section 2's observation that a physician has a usual hospital and not a usual surgeon: the median referring organization concentrates 41.9% of its patients' operations at its leading hospital but routes only 27.5% of the same referrals through its leading specialist organization, and the hospital distribution is the more concentrated of the two for 70% of organizations. And it is why the flow can carry price consequences without health consequences, a quarter of the organizational price gap travelling with the patient (Section 4), which is what a payment stream attached to a relationship rather than to measured performance looks like. Rent, in that narrow and operational sense.

Two boundaries keep the word honest. If the relationship buys coordination, whether priority, continuity or information back-flow, that value is real and nothing in this paper prices it; the health margin measured here is thirty-day surgical readmission, not everything care could protect. And the estimation identifies products of decision weight and preference, never the two separately, so no decomposition of whose utility the routing serves is on offer, and none is claimed. What the margins that can be measured say is this: reallocating inherited demand is inert on the health margin merger review never models, and consequential on the transfer margin it mismeasures. The model's silence on welfare is affordable; its error on transfers is not.

10.4 A class of markets

The structure this paper measures is not specific to hospitals. It has three conditions. An intermediary's standing habit is the consideration set: here, a referral chain that places 2.4 of 48 available hospitals in front of the patient. The intermediary outlives the relationships inside it, and its holdings pass at organizational events: here, menus that survive the replacement of the physicians who use them, in a market where only 14.6% of repeat patients keep an intact chain across four years. And the principal's stakes are episodic, so

she never accumulates standing of her own: here, 87.8% of operations are the patient's first in the six years observed, and her override of the menu rises only with the experience she rarely gets (Section 10.1). Call the structure *default-succession demand*: allocation neither by consumer preference nor by procurement, but by an intermediary's habit that passes with the intermediary.

Other markets satisfy the conditions. In retail financial advice, the adviser's own portfolio, not the client's characteristics, determines the client's allocation (Foerster et al., 2017), and advisory books of business are themselves bought and sold: mortal capital made heritable, the move of Section 10.2 in another industry. In auditing, mandated rotation of engagement partners presumes exactly what Section 3 measures: that the client attaches to the firm and survives the people. And gatekept public systems are the institutionalized version: when the English NHS freed patients to choose their hospital, consideration sets did the work the reform assigned to preferences (Gaynor, Propper and Seiler, 2016). For any member of the class the prediction is the same: market shares that outlive every relationship that produced them, entry that contests little, and models of principal choice that overstate contestability, with one sign.

Within health care the correction travels by construction. The two screens of Section 9 are not special; they share an input with the rest of the toolbox. Predicted-flow concentration indexes (Kessler and McClellan, 2000), willingness-to-pay screens (Capps, Dranove and Satterthwaite, 2003), and the literature evaluating those screens (Garmon, 2017) are built on patient-choice demand systems, so they inherit what the shared input gets wrong about substitution. What Section 9 documents is a property of that input, not of the two screens tested: displaced demand stays with the incumbent's owner more, and spreads across the map less, than a patient-choice system predicts. And part of what looks like place in geographic decompositions of care (Finkelstein, Gentzkow and Williams, 2016) is a lineage of referring organizations, carried by patients who never chose it.

Who chooses your hospital? For the median elective operation in these data, nobody living: the choice was assembled by organizations that no longer exist, transmitted through relationships that have since dissolved, and exercised by default. A market can be allocated this way and look, from above, perfectly stable. But every method that reads the stability as preference will mismeasure the market in the same direction, for as long as the assumption stands.

11 Conclusion

Hospital market shares in California barely move. Across four years spanning a pandemic, within-county shares correlate at 0.979. The natural reading is that patients have stable preferences over stable hospitals, and that a demand system estimated on their choices will recover those preferences.

This paper reports that the machinery underneath is not stable at all. Only 14.6% of repeat patients keep both their primary-care and specialist relationships intact over the same four years. Organizations dissolve, are absorbed, and change hands continuously, and at least 36% of same-service hospital switching among repeat patients is attributable to that succession rather than to anything the patient did. Stillness on top, churn underneath.

What reconciles the two is that demand is attached to a menu rather than chosen. A patient has forty-eight hospitals within reach that perform her operation; her referral chain places 2.4 of them in front of her. Nobody currently alive selects those two. Replacing a quarter of an organization's physicians does not move its hospital distribution; a physician who changes employer is two-thirds re-routed toward her new employer's hospitals within three years; and the distribution responds to no price, quality, entry or contract margin I can measure. The menu is inherited, and it changes hands when an organization dies. Shares persist because the asset is bequeathed, not because anyone keeps choosing it.

That has a specific consequence for the one calculation merger review depends on. A demand system in which patients choose returns less within-system diversion than two unrelated real shocks measure: replayed on the very episodes the first shock provides, it recovers 41% of measured within-system retention against the menu model's 88%, and both shortfalls carry confidence intervals that exclude zero. The error runs in the direction that clears combinations rather than flags them. It is not a distance story—both models send the same share of displaced demand to the nearest alternative—but a concentration story: displaced demand goes to markedly fewer places, and to different ones, and 30% of it reaches no hospital the patient would have considered.

What I am not claiming

The composite objective is not decomposed. Only products of decision weights and preferences are identified here, and the within-menu index is not a welfare measure; no dollar conversion is offered and none would be defensible. Severing a patient's chain moves where she is treated without detectably harming her—readmission is no higher on the exogenous arm, with a 95% upper bound on harm of 0.15 percentage points—but that is a statement about outcomes, not about welfare, and the sample is not powered to find small effects.

Nor does displaced demand settle where the habit lives. On realized reallocation a model carrying the menu in utility performs like the consideration model, so every screening conclusion is robust to that placement; the consideration structure is preferred on the moments only it speaks to—the countable menu, the revealed exclusions, and the fact that a patient inside one relationship repeats her hospital at rates a model redrawing shocks each episode cannot produce.

No transaction is evaluated. The candidate pairs in Section 9 are hypothetical, the

margin and threshold in the diversion screen are assumed rather than estimated, and nothing here says any particular merger was decided wrongly.

And the origin of a menu is not recovered. Six years of claims cannot observe an organization's founding. I can show that the menu is not explained by geography, by insurance networks, by the identity of the physicians, or by any supply-side margin a hospital controls—nor by its physicians' own training histories: an alumni tie between a practice's senior physicians and a hospital's staff moves routing by 0.003 (0.015), a tight zero in a design that recovers its own distance gradient. What remains is that it is historical, and history before 2018 is outside the data.

What follows for measurement

The practical implication is narrower than a policy recommendation and more useful than a caveat. The inputs that identify the menu—which organization refers a patient, and where that organization's other patients go—are already in the claims data that merger analysis uses. They are not exotic. A screening exercise that conditions diversion on the referring organization is available today, and where it has been checked against real shocks it lands closer than the model in use.

There is also a form of concentration the screens do not observe. Over 2018–2023 the referring layer's Herfindahl index rose 25.9% while the hospital layer's rose 4.0%, on the same markets and the same weights. A referrer HHI is not a market-power statistic in the sense a hospital HHI is, and the referral layer starts from a much lower base, so the comparison should not be pushed further than it goes. What it does establish is that the layer determining which hospital a patient reaches is consolidating several times faster than the layer that is monitored—and that it consolidates by succession, so there is no filing to review and no threshold to clear.

A last observation

The title asks who chooses your hospital. On the evidence here the honest answer is that nobody living does. The choice was made by a sequence of relationships that formed before the patient arrived and that will outlast the physicians who currently staff them. That is an uncomfortable object for a literature built on discrete choice, because the agent it needs is absent. It is a more tractable object than it first appears, though: an inherited menu is observable, it is stable enough to estimate, and its consequences for the quantities antitrust computes can be measured against shocks rather than assumed. What it cannot be is ignored, because a model that assumes the patient chooses will keep returning the wrong number in the same direction.

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Online Appendix

A Construction details

Episodes. An operation generates two claims: a professional claim carrying the surgeon and a facility claim carrying the building, linked on member and date rather than on a shared identifier, because inpatient facility bills carry no procedure code. The site of care (hospital outpatient department, freestanding surgery center, inpatient) comes from the facility claim's setting code. An episode's focal procedure is restricted to true operations—vascular-access and antepartum-monitoring codes, which carry surgical billing flags without being operations, are excluded.

The unit of routing. The referring organization is the billing entity, resolved from each member's calendar-year modal biller among her primary-care visits. California's corporate-practice bar means hospital-owned practices bill through foundations under their own identifiers, so ownership is resolved through Medicare enrollment reassignment and the American Hospital Association system registry rather than billing identity.

Choice sets and menus. Two candidate sets appear in the paper and they are different objects. The demand model uses the fifty nearest facilities within one hundred miles offering the patient's procedure group in her year, with the chosen facility always included. The movers design uses the union of the two organizations' own destination lists, defined in Section 4. Menus are the referring organization's distribution over destinations, built from members who never switch so that no patient enters the menu she is measured against, and measured for both organizations over the same five calendar years centered on the break; the leave-one-out menu over every member is the robustness alternative. System ownership is the modal system over a facility's panel years, ties to the smallest code, throughout the paper.

Holding residence fixed. The movers design compares a patient against herself across a change of referring organization, so a change of address would confound it. The five-digit residential ZIP code is therefore required to be identical on both sides of the break. The condition is not a formality: estimated at the adjacent link of the chain, where the thinner panel makes both cells visible, patients who did move transfer 0.617 against 0.381 for those who did not, so pooling them returns an inflated figure.

Kaiser. The integrated system is removed two ways: the demand estimation sample excludes Kaiser-paid episodes, while the reduced-form evidence excludes Kaiser delivery facilities; the two rules differ by 61,576 operations and Table 1 reports both counts.

Table 14: Episodes with and without an identifiable referring organization

	Resolved	Unresolved
Operations	3,838,612	892,385
Hospital outpatient department	0.479	0.461
Freestanding surgery center	0.423	0.381
Inpatient	0.097	0.158
HMO	0.368	0.326
PPO	0.533	0.561
Year 2018	0.163	0.153
Seven pre-registered lines	0.061	0.047
Median distance traveled (miles)	7.97	7.77
Chose the nearest capable facility	0.154	0.173
Chosen facility within the 50-nearest set	0.847	0.843
Mean log price index of the chosen facility	0.028	0.033

Notes: The universe is the funnel's step D (in-state, elective, non-Kaiser, facility resolved); the split is whether a referring primary-care organization could be attached. The final four rows are the routing behavior both groups can exhibit without an organization being observed.

B The resolution margin

The funnel's one substantive attrition is the 18.9% of otherwise-eligible operations with no identifiable referring organization—the patient has no primary-care visits the claims can attach, or attends several organizations with no dominant one. The organization is unobservable for these episodes by definition; where the patient *went* is not. Table 14 compares the two groups on everything the unresolved side can show.

Two facts organize the table. On routing conduct—distance, use of the nearest facility, presence in the choice set, the price position of the chosen facility—the groups are within two points of each other, with the unresolved marginally *more* proximity-conforming; whatever the menu does, excluding these episodes cannot be what creates the paper's routing-versus-proximity premise. On composition they differ where the mechanism says they should: resolution fails more for inpatient operations and for looser products, and the resolution rate is 82.9% versus 80.4% across HMO and PPO and essentially flat across calendar years, so no design that splits on product or year rests on differential resolution. What the unresolved episodes cannot do is enter any within-member design—a patient with no organization cannot break with one—and the paper claims nothing about them beyond this table.

Table 15: The relationship effect under every launch definition

Launch definition	Events	Coefficient	t
All first appearances	281	0.244 (0.030)	8.2
with twenty distance bins	281	0.245 (0.030)	8.1
Zero prior episodes, operating, persists	120	0.241 (0.038)	6.4
Facility billing other groups throughout	108	0.196 (0.023)	8.5
At most three groups in the year	144	0.248 (0.021)	11.8
One group in the year *(reported)*	91	0.257 (0.020)	12.8
Risk set defined on pre-launch volume	277	0.279 (0.034)	8.3

Notes: Within-launch linear probability estimates of adoption on a pre-existing relationship, with launch fixed effects and standard errors clustered on the referring organization. Organizations that already use a facility are closer to it (median fourteen miles against twenty-seven), which is why the distance-bin row matters: twenty bins absorb any shape distance takes and move the coefficient by 0.001. The frame holds 282 launches; one has an empty risk set under the estimation filters, so the full-sample rows carry 281 events.

C Diagnostics for the service-launch design

The entry experiment defines a launch as a (facility, procedure group) pair appearing for the first time with at least twenty operations, 2019–2022. Four things can go wrong with that definition; each is measurable, and none moves the estimate (Table 15).

Left censoring. The panel begins in 2018, so an early launch has little clean pre-period. *A volume floor.* The facility-service panel records nothing below three cases a year, so a launch date can post-date a facility's true first case; checked against the raw episode records, 195 of 282 launches have zero prior episodes of the group, and 85% of those observed two further years are still performing it. *New facilities.* A launch at a facility new to the data is entry of a building, not of a service line; restricting to facilities billing other groups throughout the panel is the corresponding row of the table. *A step change in what the claims record*—the failure only inspection by name caught: one academic center appears to launch thirteen groups in one year, another twenty-four. A hospital does not add twenty-four service lines at once; what moved is how much of the facility the data capture, through a payer contract, a billing-identity change or a coverage shift. 137 of 282 launches come from sixteen such facilities, and no pre-period test detects them, because these facilities are not new: they were already billing a median of fourteen episodes in the prior year, well above the recording floor. Only the implausible pattern of many simultaneous launches gives them away.

External validation was attempted and is unavailable at useful resolution: California's hospital utilization files record total surgical volume, not volume by procedure group, and cover hospitals only, while most launches are at ambulatory surgery centers.

Table 16: The pass-through under one measurement refinement at a time

	$\hat{\delta}$
Headline (window switch object, habit from never-switchers)	0.260
Decisive switches only	0.371
The plan independently confirms the switch	0.417
Both switch definitions agree on the organizations	0.302
The plan witness by product, where gatekeeping authority differs	
Health-maintenance product, plan confirms	0.505
Health-maintenance product, plan contradicts	0.226
Preferred-provider product, plan confirms	0.296
Preferred-provider product, plan contradicts	0.231
Dating the habit; the headline centers a five-year window on the break	
Pooled over the whole panel	0.261
Matched to the break year alone	0.227
Leave-one-out habit, pooled	0.292
Placebo: permuted referring pair	0.000
Placebo: pre-trend ($t-2 \to t-1$)	0.003
Placebo: new physician, same practice grouping	0.111

Notes: The three dating rows are all symmetric—both organizations are measured over the same span, whether that span is one year, five, or the whole panel—so they differ in how precisely each share is estimated rather than in what it measures, and the corrected estimate is nearly common to them (Section 4.2). An asymmetric dating that measures the departing organization before the break and the arriving one after is excluded: it leaks a market-wide shift into G and returns a pre-trend of 0.020 against the 0.003 below. Standard errors are clustered on the destination organization and run from 0.0006 (the permutation placebo) to 0.021 across rows. Every cell holds the patient's five-digit residential ZIP code fixed. The final placebo uses the legacy practice grouping, which calls a switch internal even when the billing organizations differ, so it is not zero; where the billing organization genuinely does not change, G is identically zero and the cell is empty.

D Robustness of the pass-through estimate

The slope form of Section 4.2 is $\hat{\delta} = 0.260$ (0.009) before the menus' noise is taken out and 0.300 (0.009) after. Table 16 isolates one measurement choice per row. Every refinement of the switch moves the estimate up, which brackets the attenuation without measuring it, and Section 4.2 measures it.

The pass-through corrected for menu noise, and what the estimator's two free constants do to it:

The candidate set does not matter

Section 4.2 reports that the transfer barely moves when the candidate set is restricted to each organization's largest destinations. It is flat over the whole margin. Against the

Table 17: The destination cap, the pool gate, and the correction

	Reliability of G	OLS	Split-sample IV
Destinations per organization entering the candidate set			
5	0.902	0.264	0.301
10 *(reported)*	0.880	0.260	0.300
20	0.866	0.258	0.301
40	0.859	0.257	0.301
Patient-years a habit pool needs before it is used			
5	0.855	0.258	0.300
10 *(reported)*	0.880	0.260	0.300
25	0.906	0.258	0.301
50	0.920	0.252	0.299

Notes: Both sweeps are run on the same break-centered habit as the headline. Reliability is the Spearman–Brown extension of the correlation between two measures of G built on disjoint halves of the never-switching pool. The instrumented column is the symmetric split-sample estimator of Section 4.2. Reliability moves as the mechanism requires—down as the candidate set admits destinations both organizations rarely use, up as the pool gate admits only well-measured menus. Across the candidate set the ordinary slope tracks it, falling as reliability falls, which is the signature of attenuation. Across the pool gate it does not, because tightening the gate removes observations as well as noise. The instrumented estimate is flat under both, at 0.299 to 0.301, which is what it should be if the correction is doing its job. Across twenty random halvings it has a standard deviation of 0.0007.

unrestricted 0.4974, taking each organization's ten largest destinations gives 0.5015, its twenty-five largest 0.4982, and its fifty largest 0.4975, while the share of episodes falling inside the union rises from 84% to 94% to 97%. The restriction costs so little because episodes outside the union are at facilities the two organizations rank almost identically, so dropping them removes as much from the gap being explained as from the transfer itself.

Four routes to the same correction

Section 4.2 reports the corrected slope. Four algebraically different estimators of the same object land between 0.285 and 0.300, so their proximity is a test rather than a restatement.

The *reliability* route splits the never-switching pool in two, builds the gap twice on disjoint halves, and reads the attenuation off their correlation: 0.785 between halves, 0.880 for the full pool by Spearman–Brown, and $0.260/0.880 = 0.295$.

The *split-sample instrument* regresses the patient's destination on one half's gap using the other half's as the instrument, stacked symmetrically so neither half is privileged (Gillen, Snowberg and Yariv, 2019). It gives 0.300 (0.009). That standard error is not conditional on the menus: resampling the 795,357 members whose episodes build them and re-estimating raises it from 0.0091 to 0.0092, about one percent, in an interval otherwise dominated by variation in where patients actually went.

The *errors-in-variables* route uses no split at all. Each menu is a multinomial draw of known size—one destination per patient-year, from a pool whose size is recorded—so the sampling variance of the gap is not a quantity to be learned but one to be computed, observation by observation. Correcting the slope with it uses every patient rather than half of them and gives 0.285. It is the most conservative of them by construction: it credits the menus with only the noise independent draws would produce, and the noise actually present is two thirds larger—reliability computed from the multinomial is 0.931 against the 0.885 the split measures on the same observations. Patients within an organization are therefore not independent draws from a single profile, which is what an organization holding physicians with habits of their own should look like, and is the same structure Section 10.2 finds from the opposite direction.

The *simulation–extrapolation* route assumes no functional form for the attenuation. It adds known noise at several levels, watches the slope decay, and extrapolates back to the noise-free point. Under classical measurement error the reciprocal of the slope is exactly linear in the added noise, so the linearity is itself the specification test: it holds with $R^2 \geq 0.9999$, and the extrapolations land on the first two routes to within 0.0002.

Two over-identifying checks hold in the data rather than being assumed. A half-pool measure of the gap, being noisier still, should return $0.300 \times 0.785 = 0.2358$ under ordinary least squares, and it returns 0.2358. And the resampling above should drag the ordinary estimate down while leaving the instrumented one alone: the ordinary draws center on 0.2386 against a predicted 0.2358, and the instrumented draws on 0.2988.

The habit pool is itself a choice, and it is the conservative one: building the menus from every member with her own episodes subtracted, rather than from members who never switch, raises the corrected estimate by a further 0.010.

Dating the exit event

Section 4.2 reports that patients whose physician left the market appear to anticipate the break, by +0.231, and that the anticipation is a dating artifact. The departure is observable in the physician's own billing: in 62.8% of these breaks he had already stopped billing before the patient's recorded break date, by a median of 78 days, so the pre-period contains post-treatment time. Re-dating the event to his last active day collapses the anticipation from 0.231 to 0.072 and the transfer from 0.446 to 0.310, with overlapping confidence intervals under both clustering levels.

The ordering across exit types settles it independently, and in the direction selection cannot produce. Anticipation is *highest* where the physician also disappears from Medicare enrollment (0.295), which is the most complete exit the outside records can show and the least foreseeable, and *lowest* for retirements consistent with the physician's own age (0.194), which is the exit a patient could actually see coming. What anticipation tracks is

therefore whether the departing practice left a successor for patients to drift toward, not whether the patient chose to leave.

The price specification

Section 4.2 reports the transfer in prices. Let $\bar{p}_r$ be the mean log price index of organization r's destinations, computed once per organization and leaving the patient's own episodes out, and let p_i^{pre} and p_i^{post} be the price index of a facility she used before and after the break. The estimating equation is

$$p_i^{\text{post}} - p_i^{\text{pre}} \;=\; \alpha + S_{\text{chain}}\big(\bar{p}_{B(i)} - \bar{p}_{A(i)}\big) + \varepsilon_i. \tag{25}$$

Prices are observed for 78.4% of the frame's episodes, 3.46 million of them, the remainder being structurally unpriced capitated care, counted rather than imputed. The estimate runs on the 119,500 pre- and post-break episode pairs of 104,844 patients whose organizational price levels resolve on both sides. It is attenuated, like the others, by measurement error in the organization-level price index. And it is *asymmetric*: pass-through is 0.369 for patients moving toward cheaper organizations and 0.213 for those moving toward dearer ones, which is not a pair of mirror images and should not be reported as one number. The price spread across organizations' destinations is wide enough for the dollar figure to matter: the ninetieth-to-tenth percentile ratio of the price index is 11.9.

E The mechanism across procedure families

Decompose a patient's return to her previous hospital into three parts: the share her new organization's own routing would deliver by chance, the excess she brings herself, and the amount lost at the break. Across all breaks these are 38.8%, 45.6% and 15.6% of a comparable non-switcher's return rate. Run the same decomposition within each of ten procedure families and the patient-side excess is above chance in every one, ranging from 1.81 times chance in ear-nose-and-throat surgery to 2.92 times in neurosurgery and spine.

The ordering is suggestive rather than identified. The families at the top—spine, general surgery, gastrointestinal surgery—are the ones in which an operation is typically the end of a long course of care at a single facility, so the patient arrives with a stock of her own. The families at the bottom—ear-nose-and-throat, obstetrics, orthopedics—are more episodic. The mechanism is general; its magnitude tracks how much prior contact the patient has had.

F The Slutsky-symmetry test

Abaluck and Adams-Prassl (2021) identify limited consideration from violations of a symmetry that full consideration must satisfy: in any additive random-utility model where every alternative is considered, the share Jacobian is the Hessian of the expected-max function, so $\partial s_j/\partial V_k = \partial s_k/\partial V_j$ however unequal j and k are. Limited consideration breaks the symmetry whenever the perturbed variable moves the menu rather than utility. The design transports to this setting because the menu variable—the referring organization's habit—moves exogenously at a break while the patient's address, insurance and diagnosis do not. For a pair with inclusion probability φ and within-menu share L,

$$\frac{\partial P_1}{\partial \varphi_2} - \frac{\partial P_2}{\partial \varphi_1} \;=\; \frac{\varphi\,(2L-1)}{(1-E)^2}, \tag{26}$$

zero only when the pair is evenly matched.

Estimating both cross-effects on 223,771 predetermined facility pairs gives a difference of $+0.014$ with a 95% interval of $[-0.004, +0.032]$: the predicted sign, not distinguishable from zero. Two controls interpret the null. A negative control simulates choices with the habit in *utility*, where symmetry is an identity, and the estimator correctly returns $+0.005$ $[-0.012, +0.022]$. A positive control simulates the paper's own consideration model, where the asymmetry exists by construction—and the estimator returns $+0.002$, inside its own interval. The test cannot recover an asymmetry it is generating: equation (26) is a two-alternative formula that overstates the pairwise effect roughly elevenfold in a choice set of forty-eight, and detecting the true magnitude would require on the order of sixty times as many movers. The null is therefore a statement about power, not about consideration, and the paper draws no verdict from it.

G Diversion conventions and inference

Ownership conventions. Section 9 resolves a facility's system as the modal system over its panel years. The referrer-death series was originally computed matching systems year by year; Table 18 shows both, with member-bootstrap confidence intervals, so the sensitivity of the measured series to the convention is on the record. The modal rule adds breaks (a modal map covers origin facilities a single-year match can miss) and slightly lowers the ratio at wide radii.

The replay's inference. The paired comparison of Table 11 resamples members (2,000 draws) and recomputes measured and modeled retention on the same resampled breaks, so the gap's interval accounts for the common sampling variation. Parameter uncertainty is assessed by redrawing each model's coefficients from their estimated distributions (twenty draws) and simulation noise by re-running the menu simulator under different seeds (five);

Table 18: The measured referrer-death series under both ownership conventions

	Modal (reported)	Year-by-year
Within-system diversion	0.438	0.392
Preservation, 5 nearest	1.37 [1.30, 1.44]	1.46 [1.37, 1.54]
Preservation, 50 nearest	2.84 [2.70, 2.99]	2.96 [2.79, 3.12]

Notes: Member-block bootstrap, 2,000 draws. The facility-death witness of Section 9.1 is computed under the same modal rule, with independent facilities as their own singleton systems: 2.51 [2.12, 2.99] on 363 events, by event bootstrap with 4,000 draws.

the resulting standard deviations at the fifty-nearest radius, 0.022, 0.007 and 0.003, sit well below the 0.074 bootstrap standard error of the measured ratio at the same radius, so the sampling interval is the binding one.